BAMAT101

Multivariable Calculus and Differential Equations

Illustrated Lesson Notes — Complete Edition (Modules 1–5)

How this document is built. Every operator, method and theorem is shown *with a picture*; every "why" claim is derived from first principles and then checked against a worked example. Boxes: **Definition** (grey), **WHY this is true** (blue), **Fixing a Common Misconception** (orange), **Worked Example** (green) and **Connection** (purple, marks where an earlier idea returns).

Reading order and numbering. Modules appear in syllabus order. Section numbers (1–31) are fixed labels used in cross-references: sections 1–14 come from the first edition (Modules 2, 3 and part of 4), sections 15–31 from the second (Modules 1, the rest of 4, and 5); "Part I" and "Part II" in the text refer to these. Reference text: Thomas' Calculus, 13th ed.

Module	Topics	Pages	Sections
1 Functions of Several Variables	Partial derivatives, total differential, Jacobian, Taylor expansion, maxima/minima, Lagrange multipliers	3–12	15–20
2 Multiple Integrals	Double integrals, change of order, polar; triple integrals, cylindrical and spherical	13–16	13
3 Vector Calculus	Gradient, directional derivative, divergence, curl, potentials, line and surface integrals, Green, Stokes, Gauss	17–31	1–12
4 Ordinary Differential Equations	Constant coefficients, undetermined coefficients, variation of parameters, Cauchy–Euler, Cauchy–Legendre, applications	32–41	14, 21–24
5 Partial Differential Equations	Formation, $F(p,q)$, $F(z,p,q)$, separable, Clairaut, Lagrange, separation of variables	42–51	25–31
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Module 1 — Functions of Several Variables

15 Functions of Two Variables and Partial Derivatives

Definition

For $z = f(x, y)$, the **partial derivative** with respect to x is

$$\frac{\partial f}{\partial x} = \lim_{h \to 0} \frac{f(x + h, y) - f(x, y)}{h},$$

the ordinary derivative in which y is frozen. Likewise $\partial f/\partial y$ freezes x.

WHY this is true

What a partial derivative really is. The graph of $z = f(x, y)$ is a surface. Freezing $y = y_0$ means slicing the surface with the vertical plane $y = y_0$; the slice is an ordinary curve $z = f(x, y_0)$ in one variable. The partial derivative $f_x(x_0, y_0)$ is the *slope of that slice* at x_0. So "freeze the other variable" is not a trick — it is literally the geometric act of cutting the surface by a plane and measuring the slope of the cut.

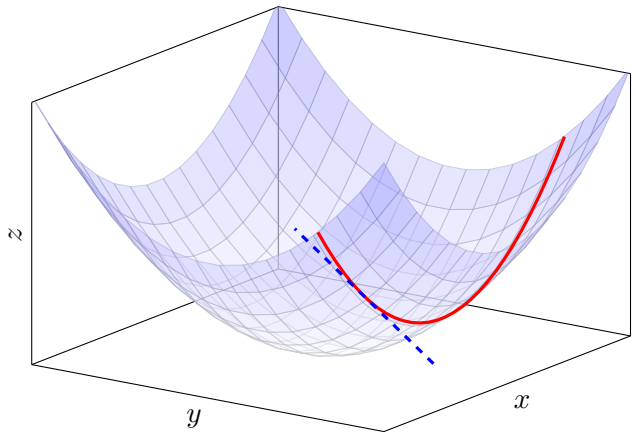


Figure 1: The surface $z = x^2 + y^2$ sliced by the plane $y = 1$. The red curve is the slice $z = x^2 + 1$; the blue dashed line is its tangent at $x = 1$. Its slope, $f_x(1, 1) = 2$, is the partial derivative.

Fixing a Common Misconception

"If both f_x and f_y exist at a point, f is well-behaved (continuous) there." This is false. Take $f(x, y) = \dfrac{xy}{x^2 + y^2}$ with $f(0, 0) = 0$. On both axes $f = 0$, so $f_x(0, 0) = f_y(0, 0) = 0$ — both exist. But along the diagonal $y = x$, $f = \frac{1}{2}$ for every $x \neq 0$, so f jumps to $\frac{1}{2}$ as you approach the origin that way: f is not even continuous at $(0, 0)$. The reason is exactly the geometric picture above: partial derivatives only inspect the surface along the *two axis slices*; the surface can misbehave in every other direction. (Section 16 shows what *is* needed for good behaviour.)

Chain rule

If $z = f(x, y)$ and $x = x(t),\ y = y(t)$, then

$$\frac{dz}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}.$$

WHY this is true

As t changes by Δt, x changes by $\Delta x \approx x'\Delta t$ and y by $\Delta y \approx y'\Delta t$. To first order, z changes by the sum of the two separate effects: $f_x\Delta x$ (the effect of moving in x) plus $f_y\Delta y$ (the effect of moving in y). Dividing by Δt and letting $\Delta t \to 0$ gives the formula. This is the same chain rule used in Part I (Section 2) to prove that ∇f is perpendicular to level surfaces; the careful justification of "to first order" is the content of Section 16.

Worked Example

$z = x^2y$ with $x = t^2,\ y = t^3$. Chain rule: $z_x = 2xy = 2t^5$, $x' = 2t$, $z_y = x^2 = t^4$, $y' = 3t^2$, so

$$\frac{dz}{dt} = 2t^5 \cdot 2t + t^4 \cdot 3t^2 = 7t^6.$$

Check by substituting first: $z = t^4 \cdot t^3 = t^7$, so $dz/dt = 7t^6$. ✓

Higher and mixed partial derivatives

Differentiating twice gives f_{xx}, f_{yy} and the *mixed* partials $f_{xy} = \partial_y(f_x)$, $f_{yx} = \partial_x(f_y)$.

WHY this is true

When is $f_{xy} = f_{yx}$? Whenever both mixed partials are continuous near the point (Clairaut–Schwarz theorem). Intuition: f_{xy} measures how the x-slope changes as you step in y, and f_{yx} how the y-slope changes as you step in x; both describe the same "twist" of the surface, computed by going around a tiny rectangle in two orders. Continuity guarantees the two orders agree. Without continuity they can differ: for $f = \dfrac{xy(x^2-y^2)}{x^2+y^2}$, $f(0,0) = 0$, one finds $f_{xy}(0,0) = -1$ but $f_{yx}(0,0) = +1$.

Worked Example

$f = x^3y^2 + \sin(xy)$. Then $f_x = 3x^2y^2 + y\cos(xy)$, and differentiating in y:

$$f_{xy} = 6x^2y + \cos(xy) - xy\sin(xy).$$

Computing in the other order, $f_y = 2x^3y + x\cos(xy)$ and $f_{yx} = 6x^2y + \cos(xy) - xy\sin(xy)$ — identical, as the theorem promises.

Laplace's equation. $u = \ln(x^2+y^2)$: $u_{xx} = \dfrac{2(y^2-x^2)}{(x^2+y^2)^2}$ and $u_{yy} = \dfrac{2(x^2-y^2)}{(x^2+y^2)^2}$, so $u_{xx}+u_{yy} = 0$. (Functions with this property model steady heat flow and electrostatic potential.)

16 The Total Differential

Definition

$$dz = \frac{\partial f}{\partial x}\,dx + \frac{\partial f}{\partial y}\,dy$$

It approximates the true change $\Delta z = f(x+dx,\, y+dy) - f(x,y)$ when dx, dy are small.

WHY this is true

Why dz approximates Δz. Move from (x, y) to $(x + dx,\, y + dy)$ in two legs: first change x (keeping $y + dy$), then change y:

$$\Delta z = \big[f(x + dx, y + dy) - f(x, y + dy)\big] + \big[f(x, y + dy) - f(x, y)\big].$$

By the one-variable mean value theorem, the first bracket equals $f_x(\xi,\, y + dy)\, dx$ and the second equals $f_y(x, \eta)\, dy$ for some intermediate ξ, η. If f_x, f_y are *continuous*, then as $dx, dy \to 0$ these evaluate to $f_x(x, y)$ and $f_y(x, y)$, giving

$$\Delta z = f_x\, dx + f_y\, dy + (\text{error that shrinks faster than } dx, dy).$$

So the requirement for dz to be a good approximation is continuity of the partial derivatives — this is what the example $\frac{xy}{x^2+y^2}$ in Section 15 lacked.

Geometric meaning. dz is the change measured along the *tangent plane* instead of along the surface. The one-variable analogue below shows the same thing: dy is the rise of the tangent line, Δy the rise of the curve, and the gap between them is the error.

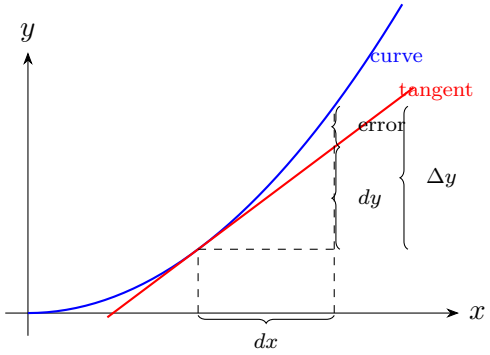


Figure 2: One-variable analogue. The tangent line's rise dy is a linear estimate of the curve's true rise Δy; the difference is the error, which shrinks much faster than dx.

Fixing a Common Misconception

"dz is the exact change." It is only the *linear estimate*. For $f = x^2 + y^2$ moving from $(1, 1)$ to $(3, 3)$: $dz = 2(1)(2) + 2(1)(2) = 8$, but the true change is $18 - 2 = 16$. The estimate is trustworthy only for small steps. (Check: $\Delta z - dz = dx^2 + dy^2 = 8$, the second-order error.)

Worked Example

Error propagation. A cylinder has $r = 5$ cm, $h = 12$ cm, each measured with possible error 0.1 cm and 0.2 cm. Estimate the maximum error in $V = \pi r^2 h$.

$$dV = 2\pi rh\, dr + \pi r^2\, dh = 2\pi(5)(12)(0.1) + \pi(25)(0.2) = 12\pi + 5\pi = 17\pi \approx 53.4 \text{ cm}^3.$$

Relative error $= \dfrac{17\pi}{300\pi} \approx 5.7\%$ — noticeably larger than either input's relative error (2% and 1.7%), because the errors accumulate.

Numerical estimate. $\sqrt{(4.02)^2 + (2.97)^2}$: take $f = \sqrt{x^2 + y^2}$ at $(4, 3)$, $f = 5$, $f_x = \frac{4}{5}$, $f_y = \frac{3}{5}$, $dx = 0.02$, $dy = -0.03$: $df = 0.016 - 0.018 = -0.002$, so the value is ≈ 4.998 (exact: 4.99813).

Connection — where this reappears

$dz = f_x\,dx + f_y\,dy$ can be written $dz = \nabla f \cdot d\mathbf{r}$. Divide by the distance $|d\mathbf{r}|$ and you obtain the directional derivative formula $D_{\hat{\mathbf{a}}} f = \nabla f \cdot \hat{\mathbf{a}}$ of Part I, Section 3. The two ideas are the same statement.

17 The Jacobian

Definition

For a change of variables $x = x(u,v),\ y = y(u,v)$ the **Jacobian** is

$$J = \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} x_u & x_v \\ y_u & y_v \end{vmatrix} = x_u y_v - x_v y_u.$$

For three variables it is the corresponding 3×3 determinant.

WHY this is true

The Jacobian is an area-scaling factor. Consider a tiny rectangle in the (u,v)-plane with sides du, dv. The map sends it to a tiny region in the (x,y)-plane. Moving du in the u direction moves the image point by $\mathbf{r}_u\,du$ where $\mathbf{r}_u = (x_u, y_u)$; moving dv moves it by $\mathbf{r}_v\,dv$ with $\mathbf{r}_v = (x_v, y_v)$. To first order the image is the *parallelogram* spanned by these two vectors, whose area is the magnitude of their cross product:

$$\text{area} = |\mathbf{r}_u \times \mathbf{r}_v|\,du\,dv = |x_u y_v - x_v y_u|\,du\,dv = |J|\,du\,dv.$$

So $|J|$ says by what factor the map stretches areas at that point. This is exactly why changing variables in a double integral puts a factor $|J|$ in: $\iint f\,dx\,dy = \iint f\,|J|\,du\,dv$.

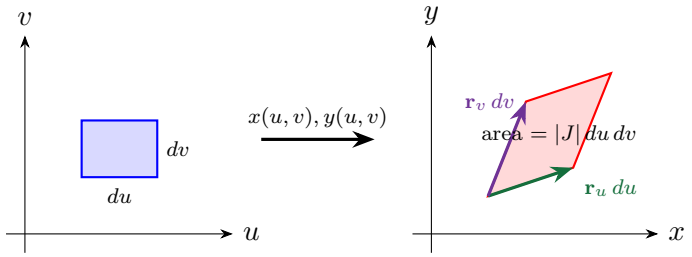


Figure 3: A small $du \times dv$ rectangle (left) is mapped to a parallelogram spanned by $\mathbf{r}_u\,du$ and $\mathbf{r}_v\,dv$ (right). Its area is $|J|\,du\,dv$.

Properties — each one derived

WHY this is true

Reciprocal property. The derivatives of (u,v) with respect to (x,y) form the matrix inverse of the derivatives of (x,y) with respect to (u,v) (that is the chain rule: $\frac{\partial(u,v)}{\partial(x,y)} \cdot \frac{\partial(x,y)}{\partial(u,v)} = I$). Determinants multiply, and $\det I = 1$, so

$$\frac{\partial(u,v)}{\partial(x,y)} \cdot \frac{\partial(x,y)}{\partial(u,v)} = 1.$$

Chain property. The same matrix-product argument gives $\dfrac{\partial(x,y)}{\partial(r,s)} = \dfrac{\partial(x,y)}{\partial(u,v)} \cdot \dfrac{\partial(u,v)}{\partial(r,s)}$ for a composite change of variables.

Functional dependence. If $J \equiv 0$ on a region, the parallelogram collapses to a segment everywhere: the image of the map is a *curve*, not a region, so u, v are related by an equation. E.g. $u = x+y,\ v = (x+y)^2$ has $J = \begin{vmatrix} 1 & 1 \\ 2(x+y) & 2(x+y) \end{vmatrix} = 0$ and indeed $v = u^2$. (A Jacobian that vanishes only at isolated points or along a curve does *not* imply dependence — it only marks where the map fails to be locally invertible.)

Worked Example

Polar coordinates. $x = r\cos\theta,\ y = r\sin\theta$:

$$J = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = r\cos^2\theta + r\sin^2\theta = r.$$

This reproduces, by pure algebra, the factor $dA = r\,dr\,d\theta$ that Part I (Section 13) obtained from the wedge-shaped-patch picture — two independent derivations agreeing.
Spherical coordinates. $x = \rho\sin\varphi\cos\theta,\ y = \rho\sin\varphi\sin\theta,\ z = \rho\cos\varphi$. Expanding the 3×3 determinant along its third row ($z_\rho = \cos\varphi,\ z_\varphi = -\rho\sin\varphi,\ z_\theta = 0$):

$$J = \cos\varphi\cdot\rho^2\sin\varphi\cos\varphi + \rho\sin\varphi\cdot\rho\sin^2\varphi = \rho^2\sin\varphi\,(\cos^2\varphi + \sin^2\varphi) = \rho^2\sin\varphi.$$

So $dV = \rho^2\sin\varphi\,d\rho\,d\varphi\,d\theta$ — the formula Part I asked you to memorise now has a derivation.
Area of an ellipse. Substitute $x = au,\ y = bv$: $J = ab$, and the ellipse $x^2/a^2 + y^2/b^2 \le 1$ becomes the unit disk, so area $= ab\iint_{u^2+v^2\le 1} du\,dv = \pi ab$.

Connection — where this reappears

The Jacobian is the bridge between Module 1 and Module 2: every "change of variables" formula in multiple integrals is a Jacobian. It returns once more in Module 5, where Lagrange's linear PDE turns out to have coefficients P, Q, R that are themselves Jacobians (Section 30).

18 Taylor's Expansion for Two Variables

Definition

About a point (a, b), with $h = x - a,\ k = y - b$ and every derivative evaluated at (a, b):

$$f(a+h,b+k) = f + \big(hf_x + kf_y\big) + \frac{1}{2!}\big(h^2 f_{xx} + 2hk\,f_{xy} + k^2 f_{yy}\big) + \frac{1}{3!}\big(h^3 f_{xxx} + 3h^2 k\,f_{xxy} + 3hk^2 f_{xyy} + k^3 f_{yyy}\big) + \cdots$$

WHY this is true

There is no new idea here — it is the one-variable Taylor series applied along a line. Walk from (a, b) towards $(a+h, b+k)$ along the straight line $(a+th,\ b+tk)$ and define $g(t) = f(a + th,\ b + tk)$, a function of one variable. Then $f(a + h, b + k) = g(1)$, and the ordinary Maclaurin series gives

$$g(1) = g(0) + g'(0) + \tfrac{1}{2!}g''(0) + \tfrac{1}{3!}g'''(0) + \cdots .$$

Now compute the derivatives with the chain rule (Section 15). First, $g'(t) = h\,f_x + k\,f_y$. Differentiating again, applying the chain rule to f_x and f_y themselves:

$$g''(t) = h\big(hf_{xx} + kf_{xy}\big) + k\big(hf_{yx} + kf_{yy}\big) = h^2 f_{xx} + 2hk\,f_{xy} + k^2 f_{yy},$$

using $f_{xy} = f_{yx}$. Setting $t = 0$ yields exactly the terms in the box. Higher terms follow the pattern of the binomial theorem applied to the operator $(h\partial_x + k\partial_y)^n$.

Fixing a Common Misconception

"The second-degree term is $h^2 f_{xx} + k^2 f_{yy}$." The mixed term is essential. It carries the factor 2 because f_{xy} and f_{yx} each contribute one copy (the two cross terms in the product $h \cdot k$ and $k \cdot h$ above). Dropping it silently changes the approximation — and, as Section 19 shows, it is the mixed term that creates saddle points.

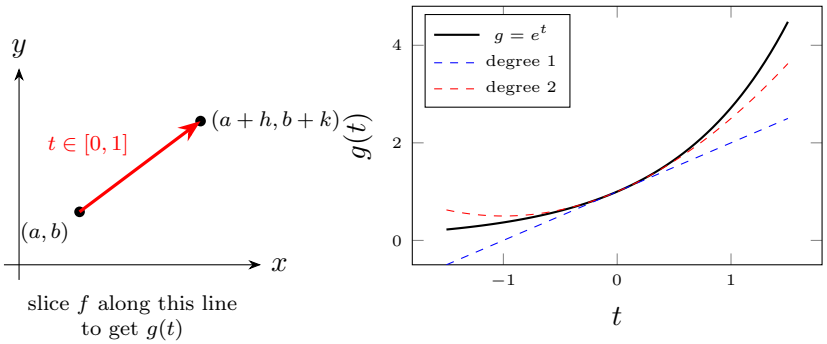


Figure 4: Left: restricting f to the line through (a, b) turns the two-variable problem into a one-variable one. Right: the one-variable Taylor polynomials hug the function more tightly as the degree rises — this is why the second-degree terms improve on the linear (tangent-plane) estimate.

Worked Example

(1) Second-degree expansion of $e^x \cos y$ about $(0, 0)$. At the origin: $f = 1,\ f_x = 1,\ f_y = 0,\ f_{xx} = 1,\ f_{xy} = 0,\ f_{yy} = -1$. Hence

$$e^x \cos y \approx 1 + x + \tfrac{1}{2}\left(x^2 - y^2\right).$$

(2) A polynomial in powers of $(x-1), (y+2)$. $f = x^2y + 3y - 2$, $h = x - 1,\ k = y + 2$. The derivatives at $(1, -2)$ are $f = -10,\ f_x = -4,\ f_y = 4,\ f_{xx} = -4,\ f_{xy} = 2,\ f_{yy} = 0,\ f_{xxy} = 2$ (others 0), so the expansion terminates:

$$f = -10 - 4h + 4k - 2h^2 + 2hk + h^2k.$$

Check: substitute $h = 0,\ k = 0$ to recover $f(1, -2) = -10$; substitute $x = 2, y = -2$ $(h = 1, k = 0)$: $-10 - 4 - 2 = -16$, and $f(2, -2) = 4(-2) - 6 - 2 = -16$. ✓

(3) Numerical approximation of $(1.1)^{1.02}$. Use $f = x^y$ about $(1, 1)$: $f = 1,\ f_x = 1,\ f_y = 0,\ f_{xx} = 0,\ f_{xy} = 1,\ f_{yy} = 0$. So $f \approx 1 + h + hk$ with $h = 0.1,\ k = 0.02$: $1 + 0.1 + 0.002 = 1.102$ (exact: 1.10210).

(4) A shortcut. For $f = \ln(1 + x + y)$, instead of computing six derivatives, put $t = x + y$ in the known series $\ln(1 + t) = t - \frac{t^2}{2} + \cdots$ to get $(x + y) - \frac{1}{2}(x + y)^2$; the derivative method gives the same result. Whenever f depends on one combination of x and y, substitution is far quicker.

Connection — where this reappears

The first-degree part, $f + hf_x + kf_y$, is the tangent-plane approximation — precisely the total differential of Section 16. The second-degree part is what Section 19 uses to decide between maximum, minimum and saddle.

19 Maxima and Minima

Definition

A point (a, b) is a **critical point** of f if $f_x = f_y = 0$ there. Let $r = f_{xx},\ s = f_{xy},\ t = f_{yy}$ at the point. Then

$rt - s^2 > 0,\ r > 0$	local **minimum**
$rt - s^2 > 0,\ r < 0$	local **maximum**
$rt - s^2 < 0$	**saddle point**
$rt - s^2 = 0$	test inconclusive

Fixing a Common Misconception

"$f_x = f_y = 0$ means a maximum or minimum." It only means the tangent plane is horizontal. For $f = x^2 - y^2$ the origin has $f_x = f_y = 0$, yet f rises along the x-axis and falls along the y-axis: a saddle. Also, checking *one* direction proves nothing — a point is a minimum only if f does not decrease in *any* direction.

WHY this is true

Where the test comes from. At a critical point the first-order terms of the Taylor expansion (Section 18) vanish, so the sign of $f(a+h, b+k) - f(a, b)$ is governed by the second-degree terms:

$$\Delta f \approx \tfrac{1}{2}\big(rh^2 + 2s\,hk + t\,k^2\big).$$

Complete the square (for $r \neq 0$):

$$rh^2 + 2s\,hk + tk^2 = r\Big(h + \frac{s}{r}k\Big)^2 + \frac{rt - s^2}{r}\,k^2.$$

- If $rt - s^2 > 0$, the two coefficients r and $\frac{rt-s^2}{r}$ have the *same* sign (that of r), so Δf has one sign in every direction: minimum if $r > 0$, maximum if $r < 0$.
- If $rt - s^2 < 0$, the coefficients have *opposite* signs, so Δf is positive in some directions and negative in others: a saddle.
- If $rt - s^2 = 0$, one direction has zero second-order change, and the third-degree terms must decide — the test says nothing.

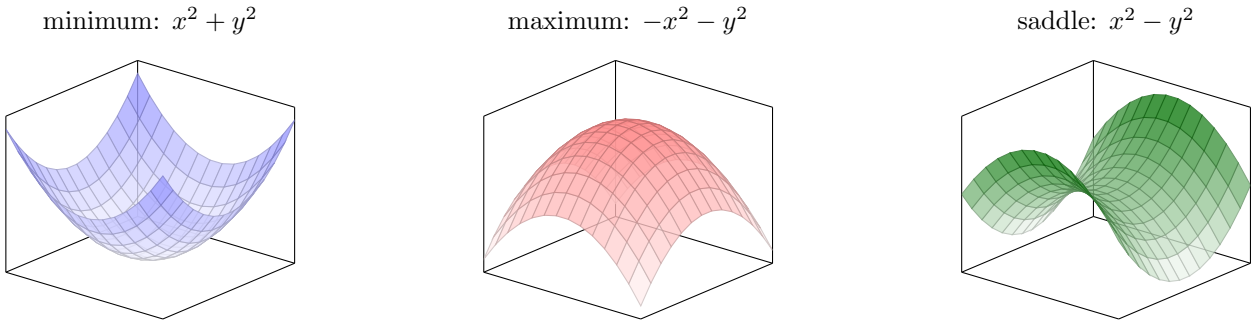


Figure 5: The three possible shapes at a critical point with $rt - s^2 \neq 0$. For $x^2 + y^2$: $r = t = 2,\ s = 0$, so $rt - s^2 = 4 > 0,\ r > 0$. For $-x^2 - y^2$: $rt - s^2 = 4 > 0,\ r < 0$. For $x^2 - y^2$: $rt - s^2 = -4 < 0$.

Worked Example

(1) Classify $f = x^3 + y^3 - 3xy$. $f_x = 3x^2 - 3y = 0,\ f_y = 3y^2 - 3x = 0$. From the first, $y = x^2$; into the second, $x^4 = x$, so $x = 0$ or $x = 1$: critical points $(0, 0), (1, 1)$. Here

$r = 6x,\ s = -3,\ t = 6y$.

- $(0,0)$: $rt - s^2 = 0 - 9 < 0$: **saddle**.
- $(1,1)$: $rt - s^2 = 36 - 9 = 27 > 0,\ r = 6 > 0$: **minimum**, $f = -1$.

(2) An application: open box of volume 32 m^3 with least surface area. Base $x \times y$, height $z = \dfrac{32}{xy}$. Area $S = xy + 2z(x+y) = xy + \dfrac{64}{y} + \dfrac{64}{x}$. Then $S_x = y - \dfrac{64}{x^2} = 0$ and $S_y = x - \dfrac{64}{y^2} = 0$ give $y = 64/x^2$ and hence $x^3 = 64$: $x = y = 4,\ z = 2,\ S_{\min} = 48$ m^2. (Second-derivative test: $S_{xx} = \frac{128}{x^3} = 2,\ S_{xy} = 1,\ S_{yy} = 2$, so $rt - s^2 = 3 > 0,\ r > 0$: a minimum.)

(3) When the test is inconclusive. $f = x^4 + y^4 - 2x^2 + 4xy - 2y^2$ has critical points $(0,0),\ (\sqrt{2}, -\sqrt{2}),\ (-\sqrt{2}, \sqrt{2})$. At the latter two, $r = t = 20,\ s = 4$: $rt - s^2 = 384 > 0$, minima with $f = -8$. At the origin $r = t = -4,\ s = 4$ gives $rt - s^2 = 0$ — inconclusive. So test directly: along $y = x$, $f = 2x^4 \geq 0$; along $y = -x$, $f = 2x^4 - 8x^2 < 0$ for small $x \neq 0$. Since f takes both signs near the origin, it is a saddle.

20 Constrained Extrema: Lagrange's Multiplier Method

Definition

To extremise $f(x, y, z)$ subject to $g(x, y, z) = 0$, solve

$$\nabla f = \lambda \nabla g \qquad \text{together with} \qquad g = 0.$$

The unknowns are x, y, z and the multiplier λ. Compare the values of f at all solutions to decide which is the maximum and which the minimum.

WHY this is true

Why the gradients must be parallel — the same argument as Part I, Section 2. Suppose P is a constrained extremum and $\mathbf{r}(t)$ is *any* curve lying on the constraint surface $g = 0$ and passing through P at $t = 0$.

1. Since $g(\mathbf{r}(t)) = 0$ for all t, the chain rule gives $\nabla g \cdot \mathbf{r}'(0) = 0$: ∇g is perpendicular to every tangent direction of the constraint surface (this is the proof that gradients are normal to level surfaces).
2. Since P is an extremum of f *among points on the surface*, the function $t \mapsto f(\mathbf{r}(t))$ has an extremum at $t = 0$, so its derivative vanishes: $\nabla f \cdot \mathbf{r}'(0) = 0$.

Thus both ∇f and ∇g are perpendicular to *every* tangent vector of the surface at P. The only direction perpendicular to a whole tangent plane is the normal direction, so $\nabla f \parallel \nabla g$, i.e. $\nabla f = \lambda \nabla g$.

Geometric picture. If a level curve of f *crossed* the constraint curve at P, you could slide along the constraint to a higher or lower value of f, so P would not be an extremum. At an extremum the level curve of f just *touches* the constraint curve — they are tangent — and tangent curves have parallel normals.

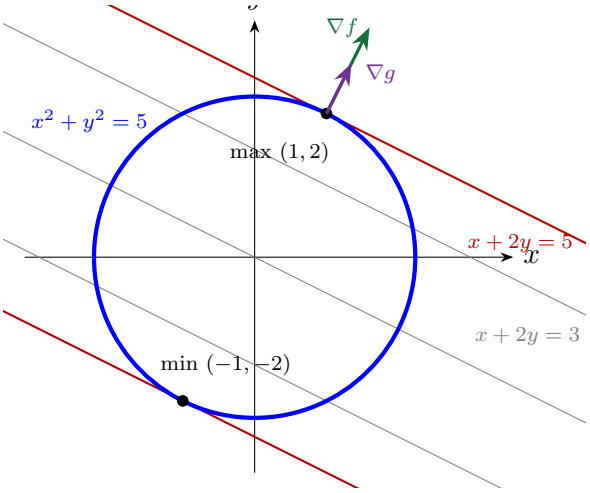


Figure 6: Maximise $f = x + 2y$ on the circle $x^2 + y^2 = 5$. The grey lines are level curves of f. The red lines $x + 2y = \pm 5$ are the extreme ones that still touch the circle — they are *tangent* to it at $(1, 2)$ and $(-1, -2)$, and there ∇f and ∇g point the same way.

Worked Example

(1) Extremes of $f = x + 2y$ on $x^2 + y^2 = 5$. $\nabla f = (1, 2)$, $\nabla g = (2x, 2y)$. $(1, 2) = \lambda(2x, 2y)$ gives $y = 2x$; the constraint becomes $5x^2 = 5$, $x = \pm 1$. Candidates $(1, 2)$ with $f = 5$ and $(-1, -2)$ with $f = -5$: maximum 5, minimum -5, exactly as the figure shows.

(2) Closest point of the plane $x + 2y + 3z = 14$ to the origin. Minimise $x^2 + y^2 + z^2$:

$(2x, 2y, 2z) = \lambda(1, 2, 3)$ gives $x = \frac{\lambda}{2}$, $y = \lambda$, $z = \frac{3\lambda}{2}$. The constraint gives $7\lambda = 14$, $\lambda = 2$: point $(1, 2, 3)$, distance $\sqrt{14}$. (Sanity check: $(1, 2, 3)$ lies in the plane and the vector $(1, 2, 3)$ is the plane's normal — the shortest path to a plane is along its normal.)

(3) Largest box in a sphere. Maximise $V = 8xyz$ on $x^2 + y^2 + z^2 = 12$. $(yz, xz, xy) = 2\lambda(x, y, z)$; multiplying the three equations by x, y, z shows $x^2 = y^2 = z^2$, so $x = y = z = 2$: a cube of edge 4, $V_{\max} = 64$.

Fixing a Common Misconception

"Lagrange's method finds the maximum." It only finds *candidates* — points where the gradients are parallel. You must still decide their nature. For $f = x^2 + y^2$ on $xy = 1$ the method gives $(1, 1)$ and $(-1, -1)$ with $f = 2$; but along the hyperbola $f = t^2 + t^{-2} \to \infty$, so 2 is the *minimum* and there is no maximum at all. Always compare values, or argue from the shape of the constraint.

13 Module 2 — Multiple Integrals

13.1 Double Integrals: Basic Evaluation

$\iint_R f(x,y)\,dA$ is evaluated as an **iterated integral**: integrate w.r.t. one variable first (holding the other fixed, exactly the reverse of a partial derivative), then integrate that result w.r.t. the second variable.

Worked Example

$\int_0^1 \int_0^2 xy^2\,dy\,dx$. Inner: $\int_0^2 xy^2\,dy = x\left[\frac{y^3}{3}\right]_0^2 = \frac{8x}{3}$. Outer: $\int_0^1 \frac{8x}{3}dx = \frac{8}{3}\cdot\frac{1}{2} = \frac{4}{3}$.

13.2 Change of Order of Integration

WHY this is true

When the inner limits depend on the outer variable, you are integrating over a *non-rectangular* region — the limits are really just a description of that region's boundary. Changing order means re-describing the *same* region with x and y's roles swapped, and the only reliable way to do that is to sketch the region first and read the new boundary curves directly off the picture. Attempting to swap limits by algebra alone, without the sketch, is the single biggest source of errors on this topic.

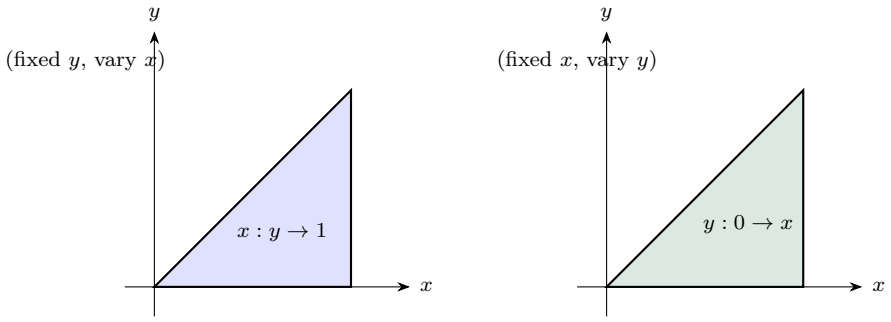


Figure 11: The same triangular region, described two ways. Left: for fixed $y \in [0,1]$, x runs from y to 1 (matches $\int_0^1\int_y^1 dx\,dy$). Right: for fixed $x \in [0,1]$, y runs from 0 to x (matches $\int_0^1\int_0^x dy\,dx$) — the flipped-order version of the exact same region.

Worked Example

Change the order in $\int_0^1 \int_y^1 f(x,y)\,dx\,dy$.

Sketch (above): the region is the triangle with vertices $(0,0),(1,0),(1,1)$, bounded by $x=y$, $x=1$, $y=0$.

Reversed: $\int_0^1 \int_0^x f(x,y)\,dy\,dx$.

13.3 Double Integrals in Polar Coordinates

Switch to polar when: the region is a circle/sector/annulus, or the integrand involves $x^2 + y^2$.

WHY this is true

$x = r\cos\theta$, $y = r\sin\theta$, and crucially $dA = r\,dr\,d\theta$ — not just $dr\,d\theta$. Here's why the extra r appears: a small "rectangle" in (r,θ)-space of size $\Delta r \times \Delta\theta$ maps to a thin wedge-shaped patch in the xy-plane. That wedge has inner arc length $r\,\Delta\theta$ and outer arc length $(r+\Delta r)\Delta\theta$, and radial thickness Δr; its area is approximately (average arc length)$\times\Delta r \approx r\,\Delta\theta\,\Delta r$. So equal steps in (r,θ) sweep out *more* actual area the farther out you are (larger r) — the factor r corrects for that stretching, exactly analogous to the $\sqrt{1+g_x^2+g_y^2}$ stretching factor for surfaces in Section 9.

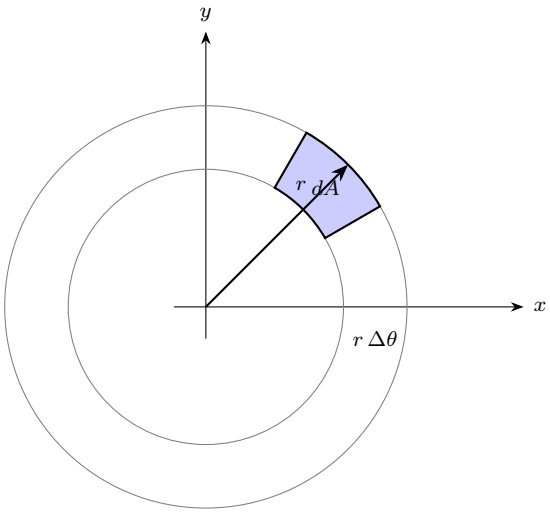


Figure 12: A polar "rectangle" of size $\Delta r \times \Delta\theta$ is really a wedge of area $\approx r\,\Delta r\,\Delta\theta$ — larger r means more area swept per unit $\Delta\theta$, which is exactly the factor $dA = r\,dr\,d\theta$ accounts for.

Worked Example

$\iint_R (x^2+y^2)\,dA$, $R : x^2+y^2 \le 4$.
Polar: $x^2+y^2 = r^2$; region becomes $0 \le r \le 2$, $0 \le \theta \le 2\pi$.

$$\int_0^{2\pi}\int_0^2 r^2 \cdot r\,dr\,d\theta = \int_0^{2\pi}\left[\tfrac{r^4}{4}\right]_0^2 d\theta = \int_0^{2\pi} 4\,d\theta = 8\pi.$$

13.4 Triple Integrals: Rectangular, Cylindrical, Spherical

Rectangular triple integrals are mechanically identical to double integrals with one more nested integral.

Worked Example

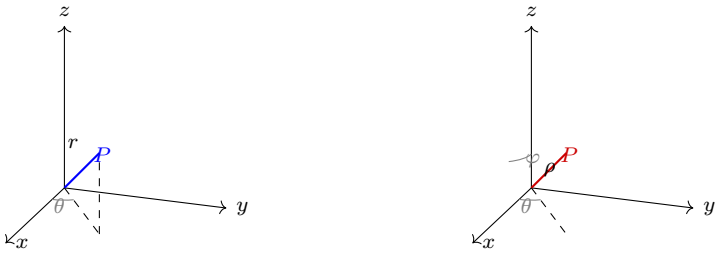
$\iiint_V xyz\,dV$ over the box $[0,1]\times[0,2]\times[0,3]$. Since the box and integrand separate cleanly:

$$\int_0^1 x\,dx \cdot \int_0^2 y\,dy \cdot \int_0^3 z\,dz = \tfrac{1}{2}\cdot 2\cdot \tfrac{9}{2} = \tfrac{9}{2}.$$

Cylindrical (circular symmetry about the z-axis): $x = r\cos\theta$, $y = r\sin\theta$, $z = z$, with $dV = r\,dr\,d\theta\,dz$ — the same r-factor as polar, since cylindrical coordinates are just polar coordinates in x, y extended along z.

Spherical (regions centered at the origin — spheres, cones): $x = \rho\sin\varphi\cos\theta$, $y = \rho\sin\varphi\sin\theta$, $z = \rho\cos\varphi$, with

$$dV = \rho^2\sin\varphi\,d\rho\,d\varphi\,d\theta.$$



Cylindrical: (r,θ,z) **Spherical:** (ρ,φ,θ)

Figure 13: Cylindrical coordinates (left): r is radial distance in the xy-plane, θ the usual angle, z unchanged. Spherical coordinates (right): ρ is distance from the origin, φ the angle down from $+z$, θ the same angle as before.

Worked Example

Cylindrical: Volume under $z = 4 - x^2 - y^2$, above $z = 0$.
Paraboloid meets $z = 0$ at $r = 2$ (since $4 - r^2 = 0$); for fixed r, θ, z runs 0 to $4 - r^2$.

$$V = \int_0^{2\pi}\int_0^2\int_0^{4-r^2} r\,dz\,dr\,d\theta = \int_0^{2\pi}\int_0^2 r(4-r^2)\,dr\,d\theta = \int_0^{2\pi}\left[2r^2 - \tfrac{r^4}{4}\right]_0^2 d\theta = \int_0^{2\pi} 4\,d\theta = 8\pi.$$

Spherical: Volume of the sphere $x^2 + y^2 + z^2 \le a^2$.

$$V = \int_0^{2\pi}\int_0^{\pi}\int_0^{a} \rho^2\sin\varphi\,d\rho\,d\varphi\,d\theta = \underbrace{\tfrac{a^3}{3}}_{\rho\text{-integral}} \cdot \underbrace{2}_{\varphi\text{-integral}} \cdot \underbrace{2\pi}_{\theta\text{-integral}} = \tfrac{4}{3}\pi a^3,$$

matching the familiar sphere-volume formula — confirming the setup is right.

System	dA or dV	Best for
Rectangular	$dx\,dy\,(dz)$	boxes, general regions
Polar	$r\,dr\,d\theta$	disks, x^2+y^2 in integrand
Cylindrical	$r\,dr\,d\theta\,dz$	cylinders, paraboloids
Spherical	$\rho^2\sin\varphi\,d\rho\,d\varphi\,d\theta$	spheres, cones from origin

Part 0 — Foundations You Need Before Module 3

Vectors, and what "bundling numbers together" means

A vector like $\mathbf{A} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$ is just three numbers (a_1, a_2, a_3) glued to three fixed, mutually perpendicular directions $\mathbf{i}, \mathbf{j}, \mathbf{k}$. This point matters more than it looks: **a vector is nothing but a triple of scalars, each one paired with a direction.** Keep this in your head — it is the exact mechanism that resolves the confusion about the gradient in Section 1.2.

Dot product and cross product

Definition

$\mathbf{A} \cdot \mathbf{B} = a_1b_1 + a_2b_2 + a_3b_3$ (a *scalar*). Geometrically $\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}||\mathbf{B}| \cos\theta$.
$\mathbf{A} \times \mathbf{B}$ (a *vector*, perpendicular to both $\mathbf{A}$ and $\mathbf{B}$), computed as the determinant

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

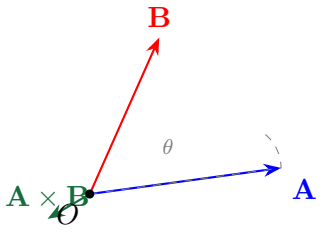


Figure 1: $\mathbf{A} \times \mathbf{B}$ is perpendicular to the plane containing $\mathbf{A}$ and $\mathbf{B}$ — direction fixed by the right-hand rule (curl fingers from $\mathbf{A}$ to $\mathbf{B}$, thumb gives the result).

Example. $\mathbf{A} = (1, 2, 0)$, $\mathbf{B} = (0, 1, 3)$.

$$\mathbf{A} \times \mathbf{B} = \mathbf{i}(2 \cdot 3 - 0 \cdot 1) - \mathbf{j}(1 \cdot 3 - 0 \cdot 0) + \mathbf{k}(1 \cdot 1 - 2 \cdot 0) = 6\mathbf{i} - 3\mathbf{j} + \mathbf{k}.$$

The $\mathbf{j}$-term always picks up a minus sign in this expansion — the single most common sign mistake in curl calculations later.

1 Scalar Fields, Vector Fields, and the Del Operator

Definition

A **scalar field** $f(x, y, z)$ assigns one *number* to every point in space (e.g. temperature). A **vector field** $\mathbf{F}(x, y, z) = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$ assigns one *vector* to every point (e.g. wind velocity).

$$\nabla = \frac{\partial}{\partial x}\mathbf{i} + \frac{\partial}{\partial y}\mathbf{j} + \frac{\partial}{\partial z}\mathbf{k}$$

∇ by itself is not a number and not a vector — it is an *instruction*: "take a partial derivative in each of the three directions." What you get out depends entirely on what you apply it to:

Notation	Operation	Input	Output
∇f	gradient	scalar field	**vector** field
$\nabla \cdot \mathbf{F}$	divergence	vector field	scalar field
$\nabla \times \mathbf{F}$	curl	vector field	vector field

2 Gradient — ∇f

Fixing a Common Misconception

"f is a scalar, so how can ∇f possibly be a vector?"
Here is the resolution, and it is exactly the point made in Part 0: ∇f is *not* f turned into a vector by some mysterious process. ∇f is *three separate scalar numbers* — $\partial f/\partial x$, $\partial f/\partial y$, $\partial f/\partial z$ — each one individually just an ordinary number (or scalar function), *bundled together as coefficients of the three perpendicular unit vectors* $\mathbf{i}, \mathbf{j}, \mathbf{k}$:

$$\nabla f = \underbrace{\frac{\partial f}{\partial x}}_{\text{scalar}}\mathbf{i} + \underbrace{\frac{\partial f}{\partial y}}_{\text{scalar}}\mathbf{j} + \underbrace{\frac{\partial f}{\partial z}}_{\text{scalar}}\mathbf{k}.$$

Compare this to how $(3, 4, 0)$ is a perfectly ordinary vector even though 3, 4, and 0 are each just plain numbers. Nothing about a number is "upgraded" into a vector; three numbers are simply reported as components *along three fixed directions at once*, and reporting three numbers-along-three-directions *is the definition of a vector*. f itself stays a scalar field the whole time — it is ∇f, a genuinely different object built *from* f's derivatives, that is a vector.

Definition

$$\nabla f = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j} + \frac{\partial f}{\partial z}\mathbf{k}$$

Worked Example

$f(x, y, z) = x^2y + yz^3$. Then $\partial f/\partial x = 2xy$, $\partial f/\partial y = x^2 + z^3$, $\partial f/\partial z = 3yz^2$.

$$\nabla f = 2xy\,\mathbf{i} + (x^2 + z^3)\,\mathbf{j} + 3yz^2\,\mathbf{k}.$$

At $(1, 2, 1)$: $\nabla f = 4\mathbf{i} + 2\mathbf{j} + 6\mathbf{k}$ — three plain numbers, reported along three directions.

Now the deeper question: WHY is ∇f perpendicular to the level surface?

This is the fact that makes the gradient useful for tangent planes and directional derivatives, so let's build it from scratch rather than accept it.

A **level surface** of f is the set of points where f equals a fixed constant c: $f(x, y, z) = c$. Picture peeling off one single "layer" of the scalar field, like one contour on a topographic map.

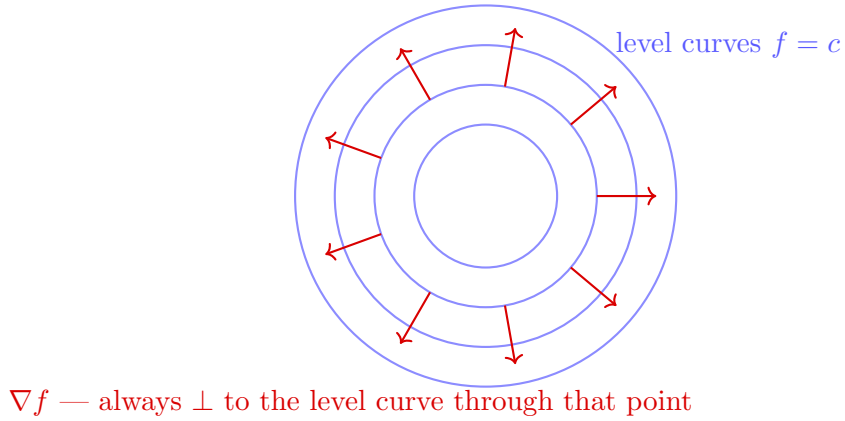


Figure 2: Level curves of $f(x, y) = x^2 + y^2$ (circles) with ∇f drawn at several points. Every arrow is perpendicular to the circle it sits on, and points toward increasing f (outward, since f grows as you move away from the origin).

WHY this is true

Claim: at any point P on the level surface $f(x, y, z) = c$, the vector ∇f is perpendicular to *every* curve that lies on that surface and passes through P.

Proof. Let $\mathbf{r}(t) = \big(x(t), y(t), z(t)\big)$ be *any* curve that lies entirely on the surface $f = c$ (i.e. $f(x(t), y(t), z(t)) = c$ for all t), and suppose it passes through P at $t = t_0$.

Since the left side is constant in t, its derivative is zero:

$$\frac{d}{dt}\Big[f\big(x(t), y(t), z(t)\big)\Big] = 0.$$

But by the multivariable chain rule, the left side expands to

$$\frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt} + \frac{\partial f}{\partial z}\frac{dz}{dt} \;=\; \nabla f \cdot \mathbf{r}'(t).$$

So $\nabla f \cdot \mathbf{r}'(t) = 0$ at every point of the curve, in particular at P. Since $\mathbf{r}'(t_0)$ is a *tangent vector to the surface at P* (it points along a curve lying in the surface), and this argument works for *any* such curve through P, the vector ∇f is perpendicular to every possible tangent direction at P — which is exactly what it means to be normal (perpendicular) to the surface at P. ■

This is the whole engine behind tangent planes, directional derivatives, and the "steepest ascent" fact below — it all comes from this one chain-rule argument.

Worked Example

Numeric check. $f(x,y) = x^2 + y^2$, level curve $f = 4$ (circle of radius 2). Take the curve on this circle: $\mathbf{r}(t) = (2\cos t,\, 2\sin t)$, which indeed satisfies $f(\mathbf{r}(t)) = 4\cos^2 t + 4\sin^2 t = 4$ for all t.
$\nabla f = 2x\,\mathbf{i} + 2y\,\mathbf{j}$. At $t = 0$ (point $(2,0)$): $\nabla f = 4\mathbf{i}$.
Tangent vector: $\mathbf{r}'(t) = (-2\sin t, 2\cos t)$, at $t = 0$: $\mathbf{r}'(0) = (0,2)$, i.e. $2\mathbf{j}$.
Check: $\nabla f \cdot \mathbf{r}'(0) = (4)(0) + (0)(2) = 0$. Perpendicular, confirmed directly by computation — not just by the abstract argument.

Application: tangent plane and normal line

If $P_0 = (x_0, y_0, z_0)$ lies on the surface $f(x,y,z) = c$, then since $\nabla f(P_0)$ is normal to the surface there:

$$\text{Tangent plane: } f_x(P_0)(x - x_0) + f_y(P_0)(y - y_0) + f_z(P_0)(z - z_0) = 0.$$

Worked Example

Find the tangent plane to $x^2 + y^2 + z^2 = 9$ at $(2,2,1)$.
$f = x^2 + y^2 + z^2$, $\nabla f = 2x\mathbf{i} + 2y\mathbf{j} + 2z\mathbf{k}$; at $(2,2,1)$: $\nabla f = 4\mathbf{i} + 4\mathbf{j} + 2\mathbf{k}$.
Tangent plane: $4(x-2) + 4(y-2) + 2(z-1) = 0 \Rightarrow 2x + 2y + z = 9$.

The other big consequence: steepest ascent

WHY this is true

Since ∇f is perpendicular to the level surface, it points in the direction where f changes *fastest* — moving *along* a level surface changes f not at all (that's what "level" means), so the direction of fastest change must be the direction most different from every tangent direction, i.e. the normal direction. Section 1.3 makes this precise with a formula.

3 Directional Derivative

Definition

The rate of change of f at a point, in the direction of a *unit* vector $\hat{\mathbf{a}}$, is

$$D_{\hat{\mathbf{a}}} f = \nabla f \cdot \hat{\mathbf{a}}.$$

WHY this is true

Why must $\hat{\mathbf{a}}$ be a unit vector? $D_{\hat{\mathbf{a}}} f$ is meant to answer "how fast does f change *per unit distance* traveled in this direction?" If $\hat{\mathbf{a}}$ had length 5, dotting with it would silently multiply the answer by 5 for no geometric reason — you'd be measuring change per 5 units of some arbitrary vector, not a comparable, standardized rate. Normalizing first is what makes $D_{\hat{\mathbf{a}}} f$ meaningful and comparable across different directions.
Why does the dot product even capture "rate of change in a direction"? Because $\nabla f \cdot \hat{\mathbf{a}} = |\nabla f| \cos\theta$, where θ is the angle between the gradient and your chosen direction (Part 0). $\cos\theta$ is largest ($= 1$) when you walk exactly along ∇f, and this is precisely the same chain-rule idea from Section 2: $D_{\hat{\mathbf{a}}} f$ is really just $\frac{d}{dt} f(\mathbf{r}(t))$ for a straight path in direction $\hat{\mathbf{a}}$, evaluated at $t = 0$.

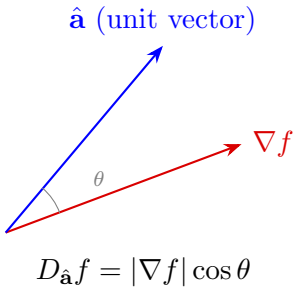


Figure 3: The directional derivative is the projection of ∇f onto the chosen direction $\hat{\mathbf{a}}$ — maximal when $\hat{\mathbf{a}}$ aligns with ∇f itself ($\theta = 0$).

Worked Example

$f(x, y, z) = xy + yz + zx$. Directional derivative at $(1, -1, 2)$ in the direction of $(1, 2, 2)$.
$\nabla f = (y + z)\mathbf{i} + (x + z)\mathbf{j} + (y + x)\mathbf{k}$; at $(1, -1, 2)$: $\nabla f = \mathbf{i} + 3\mathbf{j} + 0\mathbf{k}$.
$|(1, 2, 2)| = 3$, so $\hat{\mathbf{a}} = (1/3, 2/3, 2/3)$.
$D_{\hat{\mathbf{a}}} f = (1)(1/3) + (3)(2/3) + (0)(2/3) = 1/3 + 2 = 7/3$.

WHY this is true

Maximum and minimum rate of change. Since $D_{\hat{\mathbf{a}}} f = |\nabla f| \cos\theta$ and $|\hat{\mathbf{a}}| = 1$ is fixed, this expression is controlled entirely by $\cos\theta$, which ranges from -1 to 1. So:

- Maximum value $|\nabla f|$ occurs at $\theta = 0$, i.e. walking along ∇f itself.
- Minimum value $-|\nabla f|$ occurs at $\theta = 180°$, i.e. walking along $-\nabla f$.

This is the rigorous version of the "steepest ascent" claim from the end of Section 2.

Worked Example

$f = x^2 y$ at $(2, 1)$. $\nabla f = 2xy\,\mathbf{i} + x^2\,\mathbf{j} = 4\mathbf{i} + 4\mathbf{j}$. $|\nabla f| = \sqrt{32} = 4\sqrt{2}$.
Maximum rate of increase at $(2, 1)$ is $4\sqrt{2}$, in the direction $(1, 1)/\sqrt{2}$.

4 Divergence — $\nabla \cdot \mathbf{F}$

Definition

For $\mathbf{F} = F_1\mathbf{i} + F_2\mathbf{j} + F_3\mathbf{k}$:

$$\operatorname{div}\mathbf{F} = \nabla \cdot \mathbf{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} \quad (\text{a } \textit{scalar}).$$

WHY this is true

Why this particular combination measures "spreading out." Picture $\mathbf{F}$ as the velocity of a fluid, and imagine a tiny box of dimensions $\Delta x \times \Delta y \times \Delta z$ centered at a point. Fluid crosses the box's right face $(x+\Delta x/2)$ roughly at rate $F_1(x+\Delta x/2)\Delta y\Delta z$, and enters the left face at rate $F_1(x-\Delta x/2)\Delta y\Delta z$. The *net outflow* through this pair of faces is approximately

$$\big[F_1(x+\Delta x/2) - F_1(x-\Delta x/2)\big]\Delta y\Delta z \approx \frac{\partial F_1}{\partial x}\Delta x\Delta y\Delta z$$

by the definition of the derivative as a rate of change over a small step. Add the analogous net outflow through the other two pairs of faces ($\partial F_2/\partial y$ and $\partial F_3/\partial z$ terms) and divide by the box's volume $\Delta x\Delta y\Delta z$ to get outflow *per unit volume* — and that sum is exactly $\nabla \cdot \mathbf{F}$. So divergence is, by construction, the net outward flow rate per unit volume at a point: positive means the point acts as a source, negative means a sink.

Positive divergence (source) **Negative divergence (sink)**



Figure 4: Left: $\mathbf{F} = x\mathbf{i} + y\mathbf{j}$ (arrows flow outward everywhere $\Rightarrow \operatorname{div}\mathbf{F} > 0$, a source). Right: $\mathbf{F} = -x\mathbf{i} - y\mathbf{j}$ (arrows flow inward $\Rightarrow \operatorname{div}\mathbf{F} < 0$, a sink).

Worked Example

$\mathbf{F} = x^2\mathbf{i} + 2yz\mathbf{j} + xz^2\mathbf{k}$. $\operatorname{div}\mathbf{F} = 2x + 2z + 2xz$.

Definition

$\mathbf{F}$ is **solenoidal** if $\operatorname{div}\mathbf{F} = 0$ everywhere — flow that circulates but never accumulates (no sources or sinks anywhere).

Worked Example

Find a so that $\mathbf{F} = (x+3y)\mathbf{i} + (y-2z)\mathbf{j} + (x+az)\mathbf{k}$ is solenoidal.
$\operatorname{div}\mathbf{F} = 1 + 1 + a = 2 + a$. Set $2 + a = 0 \Rightarrow a = -2$.

5 Curl — $\nabla \times \mathbf{F}$

Definition

$$\operatorname{curl}\mathbf{F} = \nabla\times\mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ F_1 & F_2 & F_3 \end{vmatrix} = \left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}\right)\mathbf{i} - \left(\frac{\partial F_3}{\partial x} - \frac{\partial F_1}{\partial z}\right)\mathbf{j} + \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right)\mathbf{k}$$

The **j**-component always carries a minus sign, exactly as in the plain cross product (Part 0).

WHY this is true

Why this combination measures "rotation." Consider the **k**-component, $\partial F_2/\partial x - \partial F_1/\partial y$, and imagine a tiny square of side $\Delta x, \Delta y$ in the xy-plane, and compute the circulation (line integral, Section 8) of $\mathbf{F}$ counterclockwise around its four edges. The bottom edge contributes $\approx F_1(y)\Delta x$, the top edge (traversed right-to-left) contributes $\approx -F_1(y+\Delta y)\Delta x$; together these give $-[\partial F_1/\partial y]\,\Delta x\Delta y$. The right and left edges similarly contribute $+[\partial F_2/\partial x]\,\Delta x\Delta y$. Adding all four and dividing by the enclosed area $\Delta x\Delta y$ gives circulation *per unit area* $= \partial F_2/\partial x - \partial F_1/\partial y$ — exactly the **k**-component of curl. So curl at a point measures the tendency of the field to circulate around that point, and its direction (by the right-hand rule) is the axis a tiny paddle wheel dropped there would spin about.

Rotational field (curl $\neq 0$) **Irrotational field** (curl $= 0$)



Figure 5: Left: $\mathbf{F} = -y\mathbf{i} + x\mathbf{j}$ circulates around the origin — a paddle wheel dropped in would spin, so $\operatorname{curl}\mathbf{F} \neq 0$. Right: a straight, uniform flow like $\mathbf{F} = \mathbf{j}$ has no local swirl anywhere, so $\operatorname{curl}\mathbf{F} = 0$ even though the fluid is clearly moving.

Worked Example

$\mathbf{F} = xy\mathbf{i} - 2yz\mathbf{j} + 3zx\mathbf{k}$. Find $\operatorname{curl}\mathbf{F}$ at $(1,1,1)$.
i-comp: $\partial(3zx)/\partial y - \partial(-2yz)/\partial z = 0 - (-2y) = 2y$.
j-comp: $-\big[\partial(3zx)/\partial x - \partial(xy)/\partial z\big] = -[3z - 0] = -3z$.
k-comp: $\partial(-2yz)/\partial x - \partial(xy)/\partial y = 0 - x = -x$.
$\operatorname{curl}\mathbf{F} = 2y\mathbf{i} - 3z\mathbf{j} - x\mathbf{k}$. At $(1,1,1)$: $2\mathbf{i} - 3\mathbf{j} - \mathbf{k}$.

Definition

$\mathbf{F}$ is **irrotational** if $\operatorname{curl}\mathbf{F} = \mathbf{0}$ everywhere — no local swirl anywhere in the field.

Worked Example

Show $\mathbf{F} = (y^2 + 2xz^2)\mathbf{i} + 2xy\,\mathbf{j} + 2x^2z\,\mathbf{k}$ is irrotational.
$\mathbf{i}$: $\partial(2x^2z)/\partial y - \partial(2xy)/\partial z = 0 - 0 = 0$.
$\mathbf{j}$: $-[\partial(2x^2z)/\partial x - \partial(y^2 + 2xz^2)/\partial z] = -[4xz - 4xz] = 0$.
$\mathbf{k}$: $\partial(2xy)/\partial x - \partial(y^2 + 2xz^2)/\partial y = 2y - 2y = 0$.
$\operatorname{curl}\mathbf{F} = \mathbf{0}$. Confirmed irrotational.

6 Scalar and Vector Potential

WHY this is true

This is where irrotational/solenoidal fields become useful, not just classifiable. If $\operatorname{curl}\mathbf{F} = \mathbf{0}$, then a scalar function φ exists with $\mathbf{F} = \nabla\varphi$ (this is provable in general using Stokes' Theorem, Section 10: it forces every line integral of $\mathbf{F}$ between two fixed points to be path-independent, which is exactly the condition under which a potential exists). In practice, you don't need the general existence proof — you just need to know *when* to look for φ (curl is zero) and *how* to find it, shown below.

Definition

If $\operatorname{curl}\mathbf{F} = \mathbf{0}$, then $\mathbf{F} = \nabla\varphi$ for some scalar φ, called the **scalar potential**.
If $\operatorname{div}\mathbf{F} = 0$, then $\mathbf{F} = \operatorname{curl}\mathbf{A}$ for some vector $\mathbf{A}$, called the **vector potential**.

Finding φ: integrate F_1 w.r.t. x (adding an unknown function of y, z instead of a bare constant); differentiate the result w.r.t. y and match against F_2 to pin down that unknown function; repeat against F_3 for z.

Worked Example

Find φ for $\mathbf{F} = (y^2 + 2xz^2)\mathbf{i} + 2xy\mathbf{j} + 2x^2z\mathbf{k}$ (confirmed irrotational above).
Step 1. $\varphi = \int (y^2 + 2xz^2)\,dx = xy^2 + x^2z^2 + g(y,z)$.
Step 2. $\partial\varphi/\partial y = 2xy + \partial g/\partial y$. Match to $F_2 = 2xy$: forces $\partial g/\partial y = 0$, so $g = h(z)$.
Step 3. $\varphi = xy^2 + x^2z^2 + h(z)$; $\partial\varphi/\partial z = 2x^2z + h'(z)$. Match to $F_3 = 2x^2z$: forces $h'(z) = 0$, so $h = \text{const}$.

$$\varphi = xy^2 + x^2z^2 + C.$$

Check: $\nabla\varphi = (y^2 + 2xz^2)\mathbf{i} + 2xy\mathbf{j} + 2x^2z\mathbf{k} = \mathbf{F}$. ✓

7 Line Integrals

Definition

For a vector field $\mathbf{F}$ and a curve C from A to B, parametrized as $\mathbf{r}(t)$:

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)\, dt$$

represents the total **work** done by $\mathbf{F}$ moving a particle along C.

WHY this is true

Why this integral means "work." Break C into many tiny steps $\Delta\mathbf{r}_i$. Over each tiny step, the work contributed is approximately (force)·(displacement) $= \mathbf{F}(P_i) \cdot \Delta\mathbf{r}_i$ — the same dot-product idea as ordinary physics work, just over an infinitesimal straight step where $\mathbf{F}$ is roughly constant. Summing over all the little steps gives a Riemann sum $\sum \mathbf{F}(P_i) \cdot \Delta\mathbf{r}_i$, and letting the steps shrink to zero turns that sum into the integral $\int_C \mathbf{F} \cdot d\mathbf{r}$. Parametrizing by t is simply the bookkeeping device that turns the curved, multi-dimensional path into a single ordinary 1-D integral over t — this is the entire reason parametrization is the "whole game" here: it's what makes an otherwise geometric object computable with single-variable calculus.

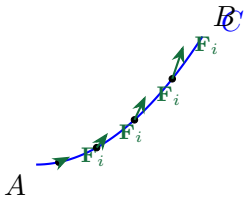


Figure 6: The curve C is chopped into small steps; at each sample point $\mathbf{F}$ is evaluated and dotted with the small displacement $\Delta\mathbf{r}_i$ along the curve. Summing (then taking the limit) gives $\int_C \mathbf{F} \cdot d\mathbf{r}$.

Worked Example

$\mathbf{F} = x\mathbf{i} + y\mathbf{j}$. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ along the *straight line* from $(0,0)$ to $(1,1)$, and separately along the *parabola* $y = x^2$ between the same two points.
Straight line: $x = t, y = t$, $t : 0 \to 1$. $\mathbf{F} \cdot d\mathbf{r} = x\,dx + y\,dy = t\,dt + t\,dt = 2t\,dt$.

$$\int_0^1 2t\,dt = 1.$$

Parabola: $x = t, y = t^2$, $t : 0 \to 1$, so $dy = 2t\,dt$. $\mathbf{F} \cdot d\mathbf{r} = t\,dt + t^2(2t\,dt) = (t + 2t^3)dt$.

$$\int_0^1 (t + 2t^3)\,dt = \left[\tfrac{t^2}{2} + \tfrac{t^4}{2}\right]_0^1 = \tfrac{1}{2} + \tfrac{1}{2} = 1.$$

Both paths give 1 — not a coincidence. It happens because this particular $\mathbf{F}$ is conservative (curl-free, with potential $\varphi = \frac{1}{2}(x^2 + y^2)$, Section 6), and Section 10's discussion of Stokes' Theorem explains *why* curl-free forces path-independence.

8 Green's Theorem in the Plane

Definition

For a closed curve C (counterclockwise) bounding a plane region R, and $\mathbf{F} = M(x,y)\mathbf{i} + N(x,y)\mathbf{j}$:

$$\oint_C (M\,dx + N\,dy) = \iint_R \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dA$$

WHY this is true

Why this is true — the cancellation argument. Chop R into a fine grid of tiny rectangles. For each small rectangle, the "local circulation" (line integral around its own tiny boundary, counterclockwise) can be computed directly and, by exactly the same face-by-face argument used for curl in Section 5, works out to $\left(\partial N/\partial x - \partial M/\partial y\right) dA$.

Now add up the circulation around *every* tiny rectangle in the grid. Look at any edge shared by two neighboring rectangles: it gets traversed once by each rectangle, but in *opposite* directions (since both rectangles go counterclockwise) — so its two contributions exactly cancel. This happens for *every internal edge*. The only edges that don't get cancelled are the ones on the outer boundary of R, which together trace out C exactly once, counterclockwise. So: (sum of all tiny local circulations) = (circulation around the outer boundary C), and simultaneously (sum of all tiny local circulations) = $\iint_R (\partial N/\partial x - \partial M/\partial y)\,dA$ by definition of the double integral as a limit of a Riemann sum. Equate the two and you have Green's Theorem.

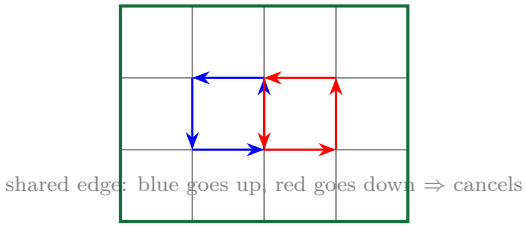


Figure 7: Two neighboring cells circulate in opposite directions along their shared edge, so that contribution cancels exactly. Doing this for every internal edge in the grid leaves only the outer boundary C contributing — the heart of Green's, Stokes', and Gauss's theorems alike.

Worked Example

Evaluate $\oint_C (xy\,dx + x^2\,dy)$, C = boundary of the unit square $[0,1]\times[0,1]$, counterclockwise. $M = xy$, $N = x^2$. $\partial N/\partial x = 2x$, $\partial M/\partial y = x$. So integrand becomes $2x - x = x$.

$$\oint_C = \iint_R x\,dA = \int_0^1\int_0^1 x\,dy\,dx = \int_0^1 x\,dx = \tfrac{1}{2}.$$

This reused the exact double-integral machinery from Module 2 — confirming that Green's Theorem is precisely where Modules 2 and 3 meet.

9 Surfaces and Surface Integrals

Definition

For a surface S with unit normal $\hat{\mathbf{n}}$ at each point, and vector field $\mathbf{F}$, the **flux integral**

$$\iint_S \mathbf{F} \cdot \hat{\mathbf{n}}\, dS$$

measures the net rate $\mathbf{F}$ "flows through" S.

WHY this is true

Why you need $\hat{\mathbf{n}}$ at all. Only the component of $\mathbf{F}$ *perpendicular* to the surface actually crosses it — a component of $\mathbf{F}$ lying flat within the surface slides along it without ever passing through. Dotting with $\hat{\mathbf{n}}$ extracts exactly that perpendicular component, which is why flux uses a dot product with the normal (rather than, say, the raw magnitude of $\mathbf{F}$).

Why $dS \neq dA$ for a tilted surface. If a surface is tilted relative to the xy-plane, a small patch of surface area dS projects down onto a *smaller* patch dA in the xy-plane — the more tilted the surface, the more the projected area shrinks relative to the true surface area. For $z = g(x, y)$, this stretching factor works out to

$$dS = \sqrt{1 + \left(\frac{\partial g}{\partial x}\right)^2 + \left(\frac{\partial g}{\partial y}\right)^2}\, dA,$$

which you can derive by noting that a small rectangle in x, y maps to a small parallelogram on the surface spanned by the tangent vectors $(1, 0, g_x)$ and $(0, 1, g_y)$, and the parallelogram's area is the magnitude of their cross product — which simplifies to exactly this square root.

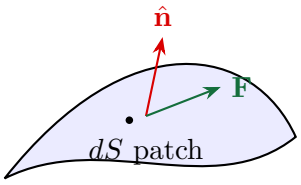


Figure 8: At a surface patch dS, only the component of $\mathbf{F}$ along the normal $\hat{\mathbf{n}}$ contributes to flux — the rest of $\mathbf{F}$ (tangential to the surface) never actually crosses it.

Worked Example

Flux of $\mathbf{F} = z\mathbf{k}$ upward through $z = 4 - x - y$ over the square $0 \le x \le 1,\ 0 \le y \le 1$.
$g = 4 - x - y$, $g_x = -1, g_y = -1$. Upward normal $\propto (1, 1, 1)$, so $\hat{\mathbf{n}} = (1, 1, 1)/\sqrt{3}$, and $dS = \sqrt{3}\, dA$.
$\mathbf{F} \cdot \hat{\mathbf{n}}\, dS = z \cdot \frac{1}{\sqrt{3}} \cdot \sqrt{3}\, dA = z\, dA = (4 - x - y)\, dA$.

$$\iint_S = \int_0^1 \int_0^1 (4 - x - y)\, dy\, dx = \int_0^1 (3.5 - x)\, dx = 3.5 - 0.5 = 3.$$

10 Stokes' Theorem

Definition

For a surface S bounded by a closed curve C (right-hand-rule orientation: curl fingers along C, thumb along $\hat{\mathbf{n}}$):

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\operatorname{curl} \mathbf{F}) \cdot \hat{\mathbf{n}}\, dS$$

WHY this is true

Stokes' Theorem is Green's Theorem lifted into 3-D. If S happens to be flat and sits in the xy-plane, then $\hat{\mathbf{n}} = \mathbf{k}$, and $(\operatorname{curl} \mathbf{F}) \cdot \mathbf{k}$ works out (Section 5) to exactly $\partial N/\partial x - \partial M/\partial y$ — so Stokes' Theorem literally *becomes* Green's Theorem in that special case. The general proof uses the same tiny-cell cancellation idea from Section 9, just applied to a curved mesh of tiny patches covering S instead of a flat grid: the local circulation around each tiny patch equals $(\operatorname{curl} \mathbf{F}) \cdot \hat{\mathbf{n}}\, dS$ for that patch (by the same argument as Section 5, run in the tangent plane of the patch), internal shared edges cancel between neighboring patches, and only the outer rim C survives.

The practical payoff: since *any* surface bounded by C gives the same answer, you are always free to pick the flattest, simplest surface stretched across C (usually a flat disk) rather than whatever oddly-shaped surface the question describes.

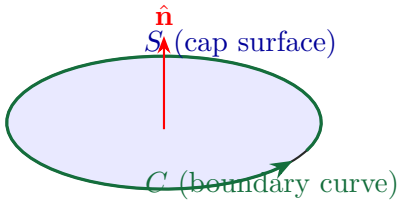


Figure 9: A curved "cap" surface S stretched across the closed curve C. Stokes' Theorem lets you swap the line integral around C for a surface integral of curl over S — and S can be *any* surface bounded by C, so pick the easiest one.

Worked Example

$\mathbf{F} = y\mathbf{i} + z\mathbf{j} + x\mathbf{k}$; C = circle $x^2 + y^2 = 1,\ z = 0$, counterclockwise viewed from $+z$.
Pick S = the flat disk $x^2 + y^2 \le 1$ in $z = 0$, so $\hat{\mathbf{n}} = \mathbf{k}$.
$\operatorname{curl} \mathbf{F}$: $\mathbf{i}$-comp $= \partial x/\partial y - \partial z/\partial z = 0 - 1 = -1$; $\mathbf{j}$-comp$= -[\partial x/\partial x - \partial y/\partial z] = -[1 - 0] = -1$;
$\mathbf{k}$-comp$= \partial z/\partial x - \partial y/\partial y = 0 - 1 = -1$.
$\operatorname{curl} \mathbf{F} = -\mathbf{i} - \mathbf{j} - \mathbf{k}$, so $(\operatorname{curl} \mathbf{F}) \cdot \mathbf{k} = -1$.

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (-1)\, dA = -(\text{area of unit disk}) = -\pi.$$

11 Volume Integrals

A volume integral $\iiint_V f\,dV$ (scalar) or $\iiint_V \mathbf{F}\,dV$ (vector, done componentwise) is mechanically identical to the triple integrals of Module 2 — no new technique. Its role here is purely as the right-hand side of the Divergence Theorem below.

12 The Gauss Divergence Theorem

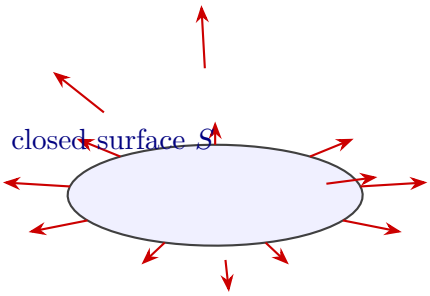
Definition

For a closed surface S enclosing a solid region V, with outward unit normal $\hat{\mathbf{n}}$:

$$\iint_S \mathbf{F}\cdot\hat{\mathbf{n}}\,dS = \iiint_V (\operatorname{div}\mathbf{F})\,dV$$

WHY this is true

This is the same cancellation idea as Green's Theorem (Section 9), one dimension further up. Chop V into a fine 3-D grid of tiny cubes. The flux out of a single tiny cube, computed directly, equals $(\operatorname{div}\mathbf{F})\,dV$ for that cube — this is exactly the box argument used to *define* divergence in Section 4, just packaged as a flux statement rather than a limit. Now sum the outward flux over *every* tiny cube in the grid. Any face shared between two neighboring cubes gets counted twice: once as "outflow" from one cube, once as "inflow" into its neighbor through the very same face — and these two contributions are equal and opposite, so they cancel. This happens at *every internal face*. Only the faces on the outer boundary S survive uncancelled. So: (sum of all tiny local outflows) = (total flux through S), and also (sum of all tiny local outflows) = $\iiint_V \operatorname{div}\mathbf{F}\,dV$ by definition of the triple integral as a limit of a Riemann sum. Equate the two.



(tiny cubes inside: shared faces cancel, only S survives)

Figure 10: Flux out through a closed surface S (arrows) equals the sum of divergence produced at every point enclosed — the Gauss Divergence Theorem, obtained from the same cell-cancellation idea as Green's Theorem, extended to three dimensions.

Worked Example

Verify the Divergence Theorem for $\mathbf{F} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ over the unit cube $[0,1]^3$.
Volume side: $\operatorname{div}\mathbf{F} = 1+1+1 = 3$. $\iiint_V 3\,dV = 3 \times 1 = 3$.
Surface side (face by face): on $x = 1$ ($\hat{\mathbf{n}} = \mathbf{i}$): $\mathbf{F}\cdot\hat{\mathbf{n}} = x = 1$, area 1, contributes 1. On $x = 0$ ($\hat{\mathbf{n}} = -\mathbf{i}$): $\mathbf{F}\cdot\hat{\mathbf{n}} = -x = 0$, contributes 0. By the same pattern, the $y = 1/y = 0$ pair contributes $1 + 0$, and the $z = 1/z = 0$ pair contributes $1 + 0$.
Total flux $= 1+0+1+0+1+0 = 3$.
Both sides equal 3. ✓ This cube/sphere/cylinder verification style is the most frequently asked question type for this theorem — practice it until it's automatic.

Module 3 — Master Summary

Operator	Input	Output	Zero-condition
∇f (gradient)	scalar field	vector field	—
$\nabla \cdot \mathbf{F}$ (divergence)	vector field	scalar	solenoidal
$\nabla \times \mathbf{F}$ (curl)	vector field	vector field	irrotational

Theorem	Converts	Requires
Green's	line integral (closed, plane) $\leftrightarrow$ double integral	2-D, closed curve
Stokes'	line integral (closed, space) $\leftrightarrow$ surface integral	3-D, uses curl
Gauss Divergence	surface integral (closed) $\leftrightarrow$ volume integral	3-D, uses divergence

14 Module 4 — Second-Order Linear ODEs

14.1 The Homogeneous Case: Complementary Function

Standard form: $ay'' + by' + cy = 0$.

WHY this is true

Why try $y = e^{mx}$? An exponential is the only kind of function that reproduces itself (up to a constant multiple) under differentiation — $\frac{d}{dx}e^{mx} = m\,e^{mx}$. That self-reproducing property is exactly what's needed to make a *linear combination* of y, y', y'' collapse to zero for every x, since every term will share the common factor e^{mx}. Substituting $y = e^{mx}$:

$$am^2e^{mx} + bme^{mx} + ce^{mx} = 0 \implies e^{mx}(am^2 + bm + c) = 0.$$

Since e^{mx} is never zero, this forces $am^2 + bm + c = 0$ — the **auxiliary equation**. The entire theory of constant-coefficient linear ODEs reduces to solving this one ordinary quadratic and reading off what kind of solution each type of root produces.

Case 1 — real, distinct roots $m_1 \neq m_2$:

$$y_{CF} = C_1e^{m_1x} + C_2e^{m_2x}.$$

Case 2 — real, equal roots $m_1 = m_2 = m$:

$$y_{CF} = (C_1 + C_2x)\,e^{mx}.$$

WHY this is true

Why the extra factor of x? With a repeated root, e^{mx} alone gives only *one* independent solution — a second-order ODE needs two. You can find the second one by treating $m_2 = m_1 + \varepsilon$ as a nearby distinct root and taking the limit $\varepsilon \to 0$ of $\dfrac{e^{(m_1+\varepsilon)x} - e^{m_1x}}{\varepsilon}$, which is precisely the definition of $\frac{\partial}{\partial m}e^{mx} = x\,e^{mx}$ evaluated at $m = m_1$. So xe^{mx} emerges naturally as the "other" solution hiding at the repeated root, and direct substitution confirms it indeed solves the ODE.

Case 3 — complex roots $m = \alpha \pm i\beta$:

$$y_{CF} = e^{\alpha x}(C_1\cos\beta x + C_2\sin\beta x).$$

WHY this is true

By Euler's formula, $e^{(\alpha+i\beta)x} = e^{\alpha x}(\cos\beta x + i\sin\beta x)$. Since the ODE is linear with real coefficients, the real part and imaginary part of this complex solution *each independently* solve the ODE — giving the two real, independent solutions $e^{\alpha x}\cos\beta x$ and $e^{\alpha x}\sin\beta x$ used above.

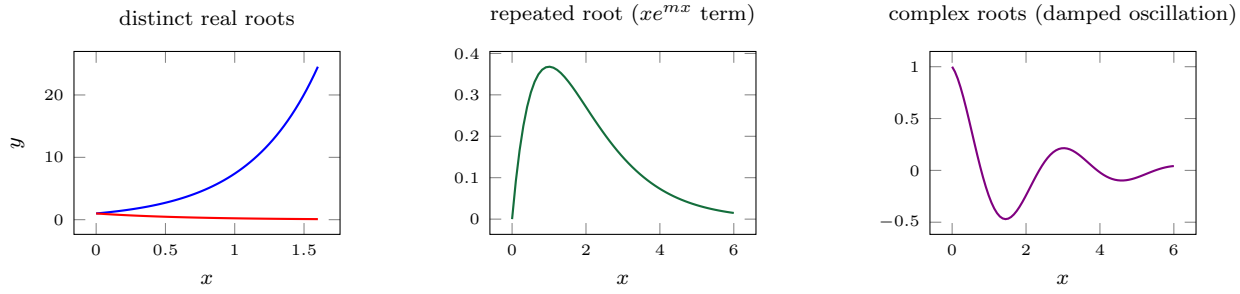


Figure 14: The shape of the solution is fixed entirely by the type of root. Left: real distinct roots give pure exponential growth/decay curves. Middle: a repeated root's second solution $x\,e^{mx}$ rises then falls even though it's built from a single exponential rate. Right: complex roots give an oscillation (cos / sin) inside an exponential envelope ($e^{\alpha x}$) — oscillating while growing or decaying.

Worked Example

$y''-5y'+6y=0$: $m^2-5m+6=0 \Rightarrow (m-2)(m-3)=0 \Rightarrow m=2,3$. $y_{CF}=C_1e^{2x}+C_2e^{3x}$.
$y''-4y'+4y=0$: $m^2-4m+4=0 \Rightarrow (m-2)^2=0 \Rightarrow m=2$ (repeated). $y_{CF}=(C_1+C_2x)e^{2x}$.
$y''+2y'+5y=0$: $m=\dfrac{-2\pm\sqrt{4-20}}{2}=-1\pm 2i$. $y_{CF}=e^{-x}(C_1\cos 2x+C_2\sin 2x)$.

14.2 Method of Undetermined Coefficients: Particular Integral

For $ay'' + by' + cy = R(x)$, the full solution is $y = y_{CF} + y_{PI}$, where y_{PI} is *any one* solution to the complete equation.

WHY this is true

The core idea. Since differentiation of e^{kx}, $\sin kx/\cos kx$, and polynomials always produces *more terms of the same family* (an exponential differentiates to an exponential, a degree-n polynomial differentiates to degree $n-1$ but its combination with lower terms still spans the same polynomial family, etc.), a linear combination of y, y', y'' built from a guess in that same family can be made to match $R(x)$ exactly by choosing the right coefficients. This is why the guess-and-match method works at all — it isn't accidental, it's guaranteed by which functions are closed under differentiation.

$R(x)$ contains	Try y_{PI} of the form
e^{kx}	$A\,e^{kx}$
$\sin kx$ or $\cos kx$	$A\cos kx + B\sin kx$
polynomial, degree n	$A_0 + A_1x + \cdots + A_nx^n$
$e^{kx}\cdot$ (poly or trig)	$e^{kx}\times$(matching guess)

Worked Example

Exponential RHS. $y'' - 5y' + 6y = e^{4x}$ (CF as above: $C_1e^{2x} + C_2e^{3x}$).
Try $y_{PI} = Ae^{4x}$: $y'_{PI} = 4Ae^{4x}$, $y''_{PI} = 16Ae^{4x}$. Substitute:

$$16Ae^{4x} - 20Ae^{4x} + 6Ae^{4x} = e^{4x} \Rightarrow 2A = 1 \Rightarrow A = \tfrac{1}{2}.$$

$$y = C_1e^{2x} + C_2e^{3x} + \tfrac{1}{2}e^{4x}.$$

Worked Example

Trigonometric RHS. $y'' + y = \cos 2x$. Auxiliary: $m^2 + 1 = 0 \Rightarrow m = \pm i$, so $y_{CF} = C_1\cos x + C_2\sin x$.
Try $y_{PI} = A\cos 2x + B\sin 2x$: $y''_{PI} = -4A\cos 2x - 4B\sin 2x$. Substitute into $y'' + y$:

$$(-4A + A)\cos 2x + (-4B + B)\sin 2x = \cos 2x \Rightarrow -3A = 1,\ -3B = 0 \Rightarrow A = -\tfrac{1}{3},\ B = 0.$$

$$y = C_1\cos x + C_2\sin x - \tfrac{1}{3}\cos 2x.$$

14.3 The Failure Case (Resonance)

WHY this is true

If the guessed y_{PI}'s functional form *already appears* in y_{CF}, substituting it produces $0 = R(x)$ — impossible, because that guess is already a solution of the *homogeneous* equation, so plugging it into the left side automatically gives zero no matter what coefficient you choose. The fix, exactly parallel to the repeated-root case in Section 14.1, is to multiply the guess by x (or by x^2 if it collides with a repeated root) — this produces a function no longer in the homogeneous solution family, breaking the automatic-zero problem while still respecting the "closed under differentiation" idea from the WHY box above.

Worked Example

$y'' - 5y' + 6y = e^{2x}$ (roots $m = 2, 3$, so $e^{2x} \in y_{CF}$ already).
Naive guess Ae^{2x} fails: substituting gives $(4A - 10A + 6A)e^{2x} = 0 \neq e^{2x}$.
Try instead $y_{PI} = Ax\,e^{2x}$: $y'_{PI} = Ae^{2x} + 2Axe^{2x}$, $y''_{PI} = 4Ae^{2x} + 4Axe^{2x}$. Substitute into $y'' - 5y' + 6y$:
xe^{2x} terms: $(4A - 10A + 6A) = 0$ — cancel as expected (confirms compatibility).
Plain e^{2x} terms: $(4A - 5A)e^{2x} = -Ae^{2x}$. Set $-A = 1 \Rightarrow A = -1$.

$$y_{PI} = -x\,e^{2x}, \qquad y = C_1e^{2x} + C_2e^{3x} - xe^{2x}.$$

If it had collided with a *repeated* root (say $m = 2$ double, so even $xe^{2x} \in y_{CF}$), you would need $y_{PI} = Ax^2e^{2x}$ instead.

14.4 Putting It Together: Initial Conditions

The general solution is always $y = y_{CF} + y_{PI}$. Apply given initial/boundary conditions *after* combining both parts.

Worked Example

Solve $y'' - 5y' + 6y = e^{4x}$, $y(0) = 0$, $y'(0) = 0$.
From before: $y = C_1e^{2x} + C_2e^{3x} + \frac{1}{2}e^{4x}$, so $y' = 2C_1e^{2x} + 3C_2e^{3x} + 2e^{4x}$.
At $x = 0$: $C_1 + C_2 + \frac{1}{2} = 0$ and $2C_1 + 3C_2 + 2 = 0$.
From the first: $C_1 = -\frac{1}{2} - C_2$. Substitute into the second: $2(-\frac{1}{2} - C_2) + 3C_2 + 2 = 0 \Rightarrow C_2 = -1$, so $C_1 = \frac{1}{2}$.

$$y = \tfrac{1}{2}e^{2x} - e^{3x} + \tfrac{1}{2}e^{4x}.$$

$R(x)$ type	Standard guess	If it collides with y_{CF}
e^{kx}	Ae^{kx}	multiply by x (or x^2 if repeated root)
$\sin kx, \cos kx$	$A\cos kx + B\sin kx$	multiply by x if same k in CF
polynomial deg n	full degree-n polynomial	shift degree up, drop redundant term
e^{kx}(trig/poly)	$e^{kx}\times$ matching guess	multiply by x under same rule

Module 4 (continued) — Ordinary Differential Equations

21 Method of Variation of Parameters

Definition

For $y'' + Py' + Qy = R(x)$ with complementary-function solutions y_1, y_2 and Wronskian $W = y_1y_2' - y_1'y_2$:

$$y_p = -y_1 \int \frac{y_2 R}{W}\,dx + y_2 \int \frac{y_1 R}{W}\,dx.$$

WHY this is true

Where the formula comes from. The CF is $c_1y_1 + c_2y_2$ with constant c_i. The idea: let the constants *vary* — seek $y_p = u_1(x)y_1 + u_2(x)y_2$. Differentiate:

$$y_p' = u_1'y_1 + u_2'y_2 + u_1y_1' + u_2y_2'.$$

We have two unknown functions u_1, u_2 but only one equation to satisfy (the ODE), so we are free to impose one extra condition. Choose the one that keeps things simple:

$$u_1'y_1 + u_2'y_2 = 0. \qquad (*)$$

Then $y_p' = u_1y_1' + u_2y_2'$ and $y_p'' = u_1'y_1' + u_2'y_2' + u_1y_1'' + u_2y_2''$. Substitute into the ODE and group by u_1 and u_2:

$$u_1 \underbrace{(y_1'' + Py_1' + Qy_1)}_{=0} + u_2 \underbrace{(y_2'' + Py_2' + Qy_2)}_{=0} + u_1'y_1' + u_2'y_2' = R.$$

The brackets vanish because y_1, y_2 solve the homogeneous equation. So we are left with a clean pair of equations in u_1', u_2':

$$u_1'y_1 + u_2'y_2 = 0, \qquad u_1'y_1' + u_2'y_2' = R.$$

Solving by Cramer's rule, with $W = y_1y_2' - y_1'y_2 \neq 0$ (independent solutions), gives $u_1' = -\dfrac{y_2R}{W}$ and $u_2' = \dfrac{y_1R}{W}$. Integrating gives the formula. The choice $(*)$ is what made the second derivatives of u_i disappear.

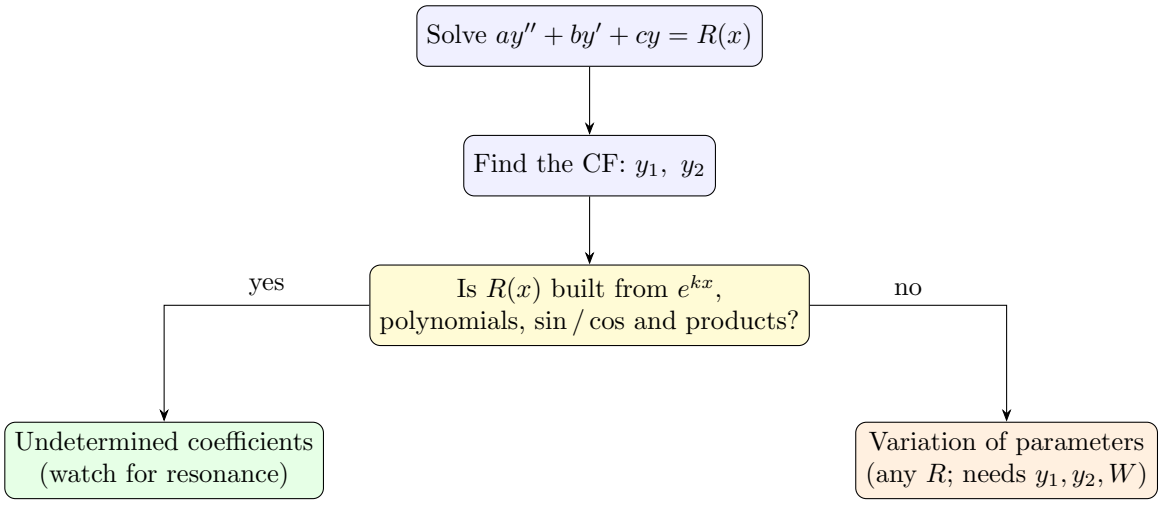


Figure 7: Choosing a method. Undetermined coefficients is shorter when it applies; variation of parameters always works once y_1, y_2 are known.

Fixing a Common Misconception

"My particular integral contains a term that is already in the CF, so it must be wrong." A particular integral is *any* one solution of the full equation — it is not unique. A CF term inside it just gets absorbed into the arbitrary constants. (See the second example: $-xe^x$ appears and is harmless.)

Worked Example

(1) $y'' + y = \sec x$**.** $y_1 = \cos x,\ y_2 = \sin x,\ W = \cos^2 x + \sin^2 x = 1$. Since $\sin x \sec x = \tan x$ and $\cos x \sec x = 1$:

$$y_p = -\cos x \int \tan x\, dx + \sin x \int dx = \cos x \ln|\cos x| + x \sin x.$$

General solution: $y = C_1 \cos x + C_2 \sin x + \cos x \ln|\cos x| + x \sin x$.

(2) $y'' - 2y' + y = \dfrac{e^x}{x}$**.** $y_1 = e^x,\ y_2 = xe^x,\ W = e^{2x}$.

$$y_p = -e^x \int \frac{xe^x \cdot e^x/x}{e^{2x}} dx + xe^x \int \frac{e^x \cdot e^x/x}{e^{2x}} dx = -e^x \int 1\, dx + xe^x \int \frac{1}{x}\, dx = -xe^x + xe^x \ln x.$$

The term $-xe^x$ merges into $C_2 xe^x$, so $y = (C_1 + C_2 x)e^x + xe^x \ln x$.

Connection — where this reappears

In Part I (Section 14) undetermined coefficients rested on the fact that derivatives of e^{kx}, polynomials and $\sin$ / $\cos$ stay inside a finite family. For $\tan x$ or $\sec x$ they do not — derivatives keep producing new functions — and that is exactly the situation this method rescues.

22 The Cauchy–Euler Equation

Definition

$$x^2 y'' + a\, xy' + b\, y = R(x) \qquad (x > 0).$$

Substitute $x = e^t$ (so $t = \ln x$) and write $D = \dfrac{d}{dt}$. Then $xy' = Dy$ and $x^2 y'' = D(D-1)y$, turning the equation into one with *constant* coefficients.

WHY this is true

Why $x = e^t$ works. With $t = \ln x$, the chain rule gives $\dfrac{dy}{dx} = \dfrac{1}{x}\dfrac{dy}{dt}$, so $x\, y' = \dfrac{dy}{dt}$. Differentiating $y' = \frac{1}{x} y_t$ once more:

$$y'' = -\frac{1}{x^2} y_t + \frac{1}{x^2} y_{tt} \implies x^2 y'' = y_{tt} - y_t = D(D-1)y.$$

Every term $x^k y^{(k)}$ becomes a constant-coefficient expression in D because the equation is *equidimensional*: replacing x by cx scales all terms equally. Multiplication in x is addition in $t = \ln x$, so "scale-invariant in x" means "translation-invariant in t" — and translation-invariant equations are exactly the constant-coefficient ones.

Shortcut. Trying $y = x^m$ directly gives $x^2y'' = m(m-1)x^m$, $xy' = mx^m$, so the homogeneous equation reduces to the *auxiliary equation*

$$m(m-1) + a\,m + b = 0.$$

Roots of $m(m-1)+am+b=0$	Solution (for $x>0$)
real, distinct $m_1 \neq m_2$	$C_1x^{m_1} + C_2x^{m_2}$
real, repeated m	$(C_1 + C_2\ln x)\,x^m$
complex $\alpha \pm i\beta$	$x^\alpha\big[C_1\cos(\beta\ln x) + C_2\sin(\beta\ln x)\big]$

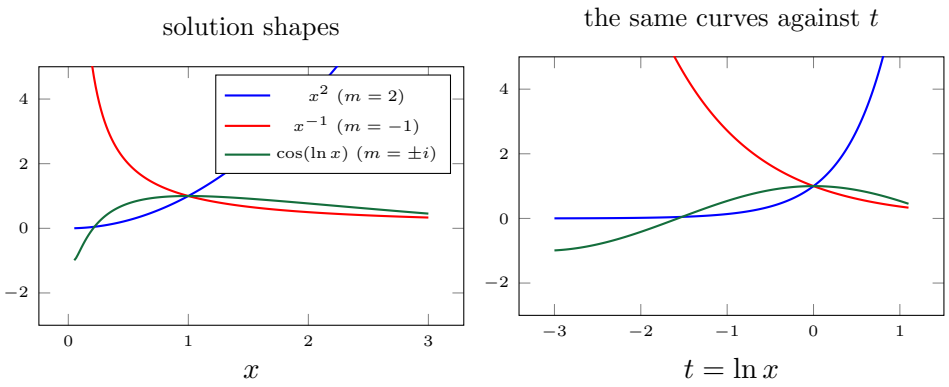


Figure 8: Left: Cauchy–Euler solutions against x. Right: the same functions against $t = \ln x$ — they become the familiar e^{2t}, e^{-t}, $\cos t$ of a constant-coefficient equation. Note that $\cos(\ln x)$ oscillates infinitely often as $x \to 0^+$, even though it never repeats on an ordinary axis.

Fixing a Common Misconception

"Try $y = e^{mx}$, as for constant-coefficient equations." That trial works only because e^{mx} keeps its shape under differentiation, leaving a common factor to cancel. With variable coefficients x^2, x there is no common factor ($x^2m^2e^{mx} + xme^{mx} + \cdots$), so nothing cancels. The natural trial is the function that is *scale-invariant*: x^m.

Worked Example

(1) $x^2y'' - 2xy' - 4y = 0$. $m(m-1) - 2m - 4 = m^2 - 3m - 4 = (m-4)(m+1) = 0$, so $y = C_1x^4 + C_2x^{-1}$.

(2) $x^2y'' + 3xy' + y = 0$. $m^2 + 2m + 1 = 0$, $m = -1$ (repeated): $y = (C_1 + C_2\ln x)\dfrac{1}{x}$.

(3) $x^2y'' + xy' + y = 0$. $m^2 + 1 = 0$, $m = \pm i$: $y = C_1\cos(\ln x) + C_2\sin(\ln x)$.

(4) Non-homogeneous with resonance: $x^2y'' - 3xy' + 4y = x^2$. With $x = e^t$: $\big[D(D-1) - 3D + 4\big]y = (D-2)^2y = e^{2t}$. CF $= (C_1 + C_2t)e^{2t}$. Since e^{2t} collides with a *repeated* root, the resonance rule of Part I (multiply by t^2) gives $y_p = \frac{1}{2}t^2e^{2t}$. Returning to x:

$$y = x^2(C_1 + C_2\ln x) + \tfrac{1}{2}x^2(\ln x)^2.$$

Connection — where this reappears

Example (4) reuses the failure-case rule of Part I (Section 14) verbatim — but in the variable t. Once $x = e^t$ is made, everything you learned about constant-coefficient equations applies unchanged.

23 The Cauchy–Legendre Equation

Definition

$$(ax+b)^2y'' + A(ax+b)y' + By = R(x).$$

Substitute $ax+b = e^t$, $D = \dfrac{d}{dt}$: then $(ax+b)y' = a\,Dy$ and $(ax+b)^2y'' = a^2D(D-1)y$. Cauchy–Euler is the special case $a=1,\ b=0$.

WHY this is true

Let $\xi = ax+b = e^t$. Then $\dfrac{dt}{dx} = \dfrac{a}{\xi}$, so $y' = \dfrac{a}{\xi}y_t$ and $\xi y' = a\,y_t$. Differentiating $y' = \frac{a}{\xi}y_t$ again, using $\frac{d\xi}{dx} = a$:

$$y'' = -\frac{a^2}{\xi^2}y_t + \frac{a^2}{\xi^2}y_{tt} \implies \xi^2y'' = a^2(y_{tt}-y_t) = a^2D(D-1)y.$$

Geometrically, the equation is a Cauchy–Euler equation *in the shifted and scaled variable* $\xi = ax+b$; the "special point" where the coefficient of y'' vanishes has moved from $x=0$ to $x=-b/a$.

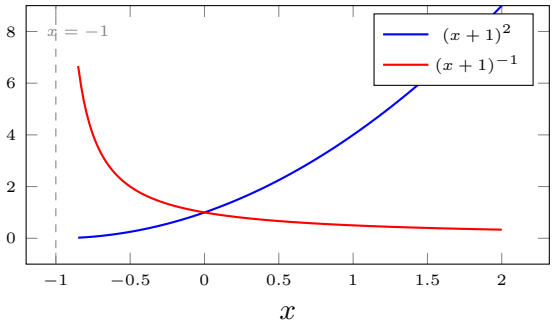


Figure 9: Solutions of $(x+1)^2y'' + (x+1)y' - y = 0$ are $(x+1)^m$ with $m = \pm 1$. The dashed line $x=-1$ plays the role that $x=0$ plays for Cauchy–Euler.

Worked Example

(1) $(3x+2)^2y'' + 3(3x+2)y' - 36y = 0$**.** Put $3x+2 = e^t$: $9D(D-1) + 3\cdot 3D - 36 = 9D^2 - 36 = 9(D^2-4)$, so $D = \pm 2$ and

$$y = C_1(3x+2)^2 + C_2(3x+2)^{-2}.$$

(2) $(2x+3)^2y'' - 2(2x+3)y' - 12y = 6x$**.** Put $2x+3 = e^t$ (so $x = \frac{e^t-3}{2}$ and $6x = 3e^t - 9$): $4D(D-1) - 2\cdot 2D - 12 = 4(D^2-2D-3)$, giving

$$(D^2 - 2D - 3)y = \tfrac{1}{4}(3e^t - 9).$$

CF: roots $3, -1$, so $C_1e^{3t} + C_2e^{-t}$. PI: for e^t, $\dfrac{3/4}{1-2-3}e^t = -\frac{3}{16}e^t$; for the constant, $\dfrac{-9/4}{-3} = \frac{3}{4}$. Back in x:

$$y = C_1(2x+3)^3 + C_2(2x+3)^{-1} - \tfrac{3}{8}x + \tfrac{3}{16}.$$

Check the particular part directly: $y_p = -\frac{3}{8}x + \frac{3}{16}$ has $y_p'' = 0$, $y_p' = -\frac{3}{8}$, so $-2(2x+3)(-\frac{3}{8}) - 12(-\frac{3}{8}x + \frac{3}{16}) = \frac{3}{4}(2x+3) + \frac{9}{2}x - \frac{9}{4} = 6x$. ✓

24 Applications of Second-Order Linear ODEs

Spring–mass–damper

WHY this is true

Newton's second law, $m\,x'' =$ (total force). The forces are the spring $-kx$ (Hooke's law, restoring), the damper $-c\,x'$ (opposing velocity) and an external force $F(t)$. So

$$m\,x'' + c\,x' + k\,x = F(t).$$

This is exactly a constant-coefficient second-order ODE, and the auxiliary equation $mr^2 + cr + k = 0$ has discriminant $c^2 - 4mk$, whose sign decides the behaviour.

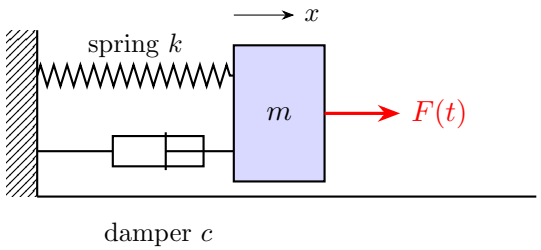


Figure 10: A mass m attached to a wall by a spring (stiffness k) and a damper (coefficient c), driven by an external force $F(t)$.

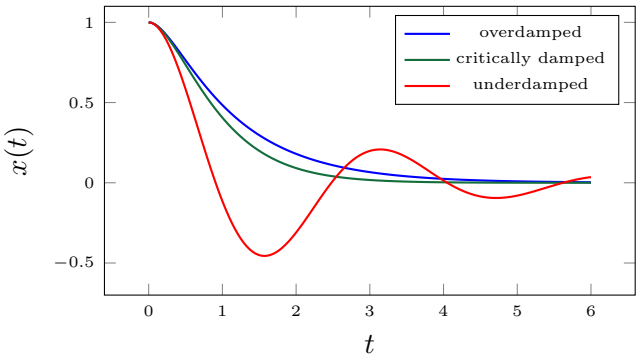


Figure 11: Three damping regimes, all released from $x(0) = 1$, $x'(0) = 0$. Overdamped ($c^2 > 4mk$): slow return, no oscillation. Critically damped ($c^2 = 4mk$): fastest return without overshoot. Underdamped ($c^2 < 4mk$): oscillation inside a decaying envelope.

Mechanical	Electrical (series LCR)	Role
mass m	inductance L	inertia
damping c	resistance R	dissipation
stiffness k	$1/C$ (elastance)	restoring
displacement x	charge q	unknown
force $F(t)$	EMF $E(t)$	driving term

Kirchhoff's voltage law (voltages around the loop sum to the EMF) gives $Lq'' + Rq' + \dfrac{q}{C} = E(t)$ — the *same* equation. Every conclusion about springs transfers to circuits, and vice versa.

Fixing a Common Misconception

"Complex roots mean the system oscillates forever." Only if there is no damping. Roots $-\alpha \pm i\beta$ with $\alpha > 0$ give $e^{-\alpha t}(\cdots)$: oscillation with a *decaying* amplitude. Persistent oscillation needs a purely imaginary pair $\pm i\beta$, i.e. $c = 0$ (or $R = 0$).

Worked Example

(1) Free vibration. A 0.5 kg mass on a spring with $k = 8$ N/m is pulled 0.3 m and released from rest; no damping. $0.5x'' + 8x = 0 \Rightarrow x'' + 16x = 0$, $\omega = 4$, so $x = 0.3\cos 4t$: amplitude 0.3 m, period $2\pi/4 = \pi/2$ s.
(2) Critical damping. $x'' + 6x' + 9x = 0$, $x(0) = 1$, $x'(0) = 0$. Repeated root -3: $x = (C_1 + C_2 t)e^{-3t}$, $C_1 = 1$, $x'(0) = C_2 - 3 = 0$, so $x = (1 + 3t)e^{-3t}$.
(3) Circuit. $L = 1$ H, $R = 4\ \Omega$, $C = \frac{1}{5}$ F, constant EMF $E = 10$ V, $q(0) = 0$, $i(0) = q'(0) = 0$. Then $q'' + 4q' + 5q = 10$ with roots $-2 \pm i$ and $q_p = 2$, so $q = e^{-2t}(A\cos t + B\sin t) + 2$. $q(0) = 0 \Rightarrow A = -2$; $q'(0) = -2A + B = 0 \Rightarrow B = -4$:

$$q = 2 - e^{-2t}(2\cos t + 4\sin t).$$

The charge settles at 2 C — the transient dies away and only the particular solution survives.

Resonance

WHY this is true

Drive an undamped system at its own natural frequency: $x'' + 4x = \cos 2t$ (natural frequency $\omega_0 = 2$). The right-hand side matches the CF $(\cos 2t, \sin 2t)$, so the failure case of Part I applies and $x_p = \dfrac{t}{4}\sin 2t$ (check: $x_p'' = \cos 2t - t\sin 2t$, so $x_p'' + 4x_p = \cos 2t$ ✓). The amplitude of x_p grows linearly with t: each push arrives exactly in phase with the motion, so energy is added every cycle and nothing removes it. With damping the growth is capped, which is why real structures survive — but badly tuned ones (the classic bridge-collapse stories) do not.

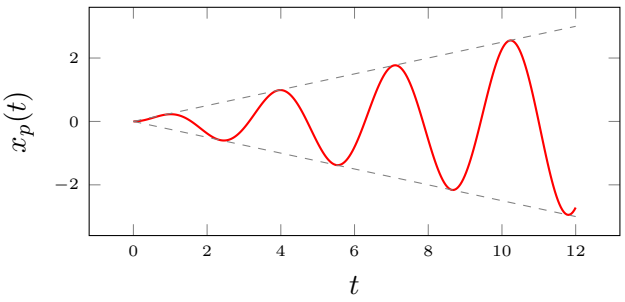


Figure 12: Resonance: $x_p = \frac{t}{4}\sin 2t$ oscillates inside the linearly growing envelope $\pm\frac{t}{4}$.

Connection — where this reappears

The failure case of Part I, Section 14 was introduced as an algebraic nuisance; here it is revealed as a physical phenomenon. The "multiply by x" rule is the mathematical signature of resonance.

Module 5 — Partial Differential Equations

Throughout, $z = z(x, y)$ and we write $p = \dfrac{\partial z}{\partial x}$, $q = \dfrac{\partial z}{\partial y}$. A first-order PDE is a relation $F(x, y, z, p, q) = 0$.

WHY this is true

The one idea behind all of Module 5. A surface $z = z(x, y)$ has, at each point (x, y, z), a tangent plane determined by the slopes (p, q). A first-order PDE is therefore a *condition on tangent planes*: "at every point, the tangent plane must have slopes (p, q) satisfying $F = 0$". Every method below is a way of building surfaces whose tangent planes obey the condition: planes themselves (Sections 26, 29), surfaces whose shape depends on a single combination of x, y (Section 27), surfaces swept out by special curves (Section 30), or products of one-variable functions (Section 31).

25 Formation of Partial Differential Equations

Definition

A PDE is **formed** by eliminating from a given family of surfaces either

- *arbitrary constants* a, b in $z = F(x, y, a, b)$, using $p = F_x$, $q = F_y$ (result: first order), or
- an *arbitrary function*, e.g. in $z = f(u(x, y))$ or $\phi(u, v) = 0$ (result: first order).

WHY this is true

Counting. From $z = F(x, y, a, b)$ we get three equations: $z = F$, $p = F_x$, $q = F_y$. They contain two unknown constants a, b; eliminating both leaves *one* relation among x, y, z, p, q — a first-order PDE. With more constants than two, first derivatives are not enough and second derivatives enter. With an arbitrary function f, differentiating gives p and q in terms of f'; dividing the two removes f' and leaves a PDE with no constants and no function: it is satisfied by *every* member of the family, and that is why a PDE has "more solutions than an ODE" — its general solution contains an arbitrary function.

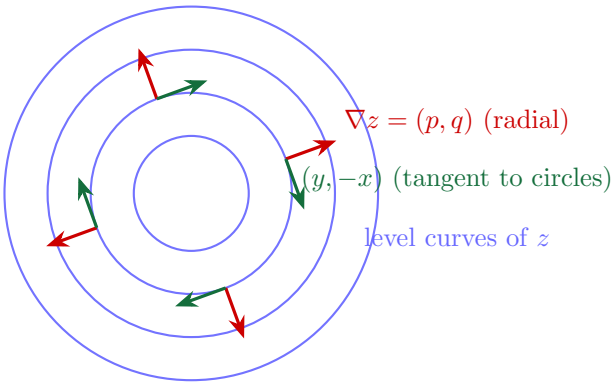


Figure 13: For $z = f(x^2 + y^2)$ the level curves are circles. The combination $yp - xq = \nabla z \cdot (y, -x)$ is the rate of change of z along the circles, which is 0: the PDE $yp - xq = 0$ says "z does not change as you go around the origin".

Worked Example

(1) Constants: $z = (x+a)(y+b)$. $p = y+b$, $q = x+a$, so $z = pq$.
(2) Constants: $z = ax+by+a^2+b^2$. $p = a$, $q = b$, so $z = px+qy+p^2+q^2$.
(3) Function: $z = f(x^2+y^2)$. $p = 2xf'$, $q = 2yf'$; dividing, $\dfrac{p}{q} = \dfrac{x}{y}$, so $yp - xq = 0$. (Figure 13 explains the meaning.)
(4) Function: $z = f(y/x)$. $p = -\dfrac{y}{x^2}f'$, $q = \dfrac{1}{x}f'$, hence $xp + yq = 0$.
(5) Two functions: $z = f(x+y) + g(x-y)$. $z_{xx} = f'' + g'' = z_{yy}$, so $\dfrac{\partial^2 z}{\partial x^2} = \dfrac{\partial^2 z}{\partial y^2}$ — the wave equation. Removing *two* arbitrary functions needs second derivatives.

General case: $\phi(u,v) = 0$, with u, v functions of x, y, z

WHY this is true

Differentiate $\phi(u,v) = 0$ with respect to x and y (remembering z depends on both):

$$\phi_u(u_x + u_z p) + \phi_v(v_x + v_z p) = 0, \qquad \phi_u(u_y + u_z q) + \phi_v(v_y + v_z q) = 0.$$

These are two linear equations for (ϕ_u, ϕ_v) which must have a non-zero solution, so the determinant vanishes:

$$(u_x + u_z p)(v_y + v_z q) - (u_y + u_z q)(v_x + v_z p) = 0.$$

Expanding (the pq terms cancel) and collecting p and q gives

$$P\,p + Q\,q = R, \quad P = \frac{\partial(u,v)}{\partial(y,z)},\ Q = \frac{\partial(u,v)}{\partial(z,x)},\ R = \frac{\partial(u,v)}{\partial(x,y)}.$$

The coefficients of the resulting PDE are *Jacobians* (Section 17).

Worked Example

$\phi(x+y+z,\ x^2+y^2+z^2) = 0$, so $u = x+y+z$, $v = x^2+y^2+z^2$. Then $P = u_y v_z - u_z v_y = 2z - 2y$, $Q = u_z v_x - u_x v_z = 2x - 2z$, $R = u_x v_y - u_y v_x = 2y - 2x$. Dividing by 2 and changing signs:

$$(y-z)\,p + (z-x)\,q = x - y.$$

(This exact equation is solved in reverse in Section 30, giving back $\phi(x+y+z,\ x^2+y^2+z^2) = 0$.)

26 Equations of the Form $F(p,q) = 0$

Definition

Method. Assume $z = ax + by + c$, so $p = a$, $q = b$. The PDE forces $F(a,b) = 0$; solve $b = \varphi(a)$. The **complete integral** is

$$z = ax + \varphi(a)\,y + c.$$

WHY this is true

Why planes work. F contains only p and q — no x, y, z. So it only constrains the *slopes* of the tangent plane, not where the plane is. A plane $z = ax+by+c$ has the *same* tangent plane (itself) everywhere, with constant slopes (a, b); it is a solution exactly when (a, b) satisfies $F(a, b) = 0$. This gives a two-parameter family (free: a and c), the complete integral.
The general integral. Choose $c = \psi(a)$ with an arbitrary ψ to get a one-parameter family of planes $z = ax+\varphi(a)y+\psi(a)$, and form its *envelope* by setting $\partial/\partial a = 0$: $0 = x+\varphi'(a)y+\psi'(a)$, then eliminate a. The envelope is again a solution because at each of its points it *touches* one of the planes, sharing that plane's position and slopes (p, q), and the PDE depends only on position and slopes.

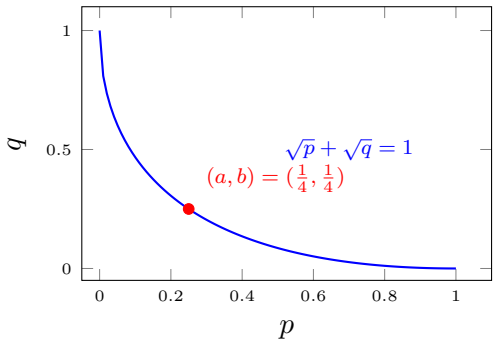


Figure 14: The "slope plane": every point (a, b) on the curve $F(p, q) = 0$ gives a solution plane $z = ax + by + c$. Here $F = \sqrt{p} + \sqrt{q} - 1$ and $b = (1 - \sqrt{a})^2$.

Fixing a Common Misconception

"The only solutions are the planes $z = ax + by + c$." Planes give only the *complete* integral. For $p^2+q^2 = 1$ the cone $z = \sqrt{x^2 + y^2}$ also works ($p = x/r$, $q = y/r$, so $p^2+q^2 = 1$), yet it is not a plane; it is the envelope of the planes $z = x\cos\alpha + y\sin\alpha$ (take $c = 0$ and eliminate α), i.e. it comes from the general integral.

Worked Example

(1) $p + q = pq$. $a + b = ab \Rightarrow b = \dfrac{a}{a-1}$. Complete integral: $z = ax + \dfrac{a}{a-1}y + c$.
(2) $\sqrt{p} + \sqrt{q} = 1$. $b = (1 - \sqrt{a})^2$ (for $0 \le a \le 1$): $z = ax + (1 - \sqrt{a})^2 y + c$.
(3) $q = p^2$. $b = a^2$: $z = ax + a^2y + c$. *General integral with $c = 0$: $\partial/\partial a$* gives $x + 2ay = 0$, so $a = -\dfrac{x}{2y}$ and $z = -\dfrac{x^2}{2y} + \dfrac{x^2}{4y} = -\dfrac{x^2}{4y}$. **Check:** $p = -\dfrac{x}{2y}$, $q = \dfrac{x^2}{4y^2} = p^2$. ✓

27 Equations of the Form $F(z, p, q) = 0$

Definition

Method. Put $u = x + ay$ and look for $z = f(u)$. Then $p = \dfrac{dz}{du}$, $q = a\dfrac{dz}{du}$, and the PDE becomes an ODE in z and u. Solve it and replace u by $x + ay$.

WHY this is true

Why this substitution. Because $F(z, p, q) = 0$ contains no explicit x or y, shifting the whole picture sideways in the xy-plane maps solutions to solutions (translation invariance). So we

can look for solutions that are constant along a whole family of parallel lines $x + ay = \text{const}$ — surfaces that depend on position only through $u = x + ay$. For such a surface, $z = f(u)$ gives $p = f'(u) \cdot 1$ and $q = f'(u) \cdot a$ by the chain rule. The parameter a (the direction of the lines) is the free constant, and integrating the ODE introduces a second one: a complete integral.

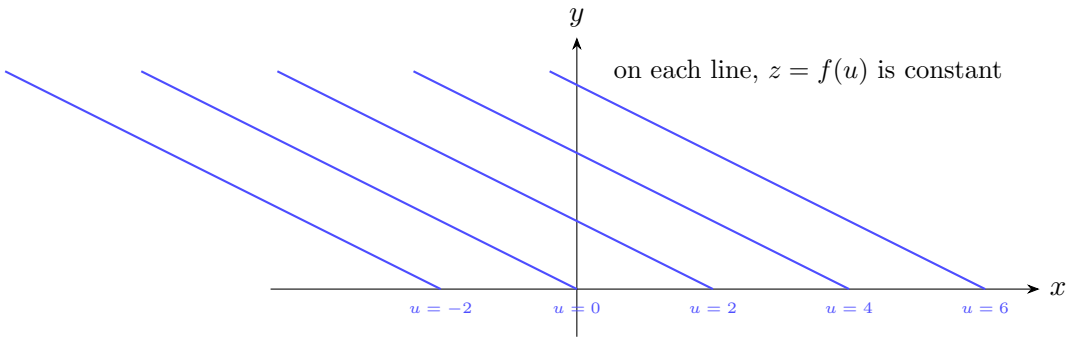


Figure 15: For $u = x + 2y$, z is constant on each line $x + 2y = \text{const}$. The surface is a "travelling profile" $z = f(u)$ stretched along the lines. This one-variable reduction is what turns the PDE into an ODE.

Fixing a Common Misconception

Using $u = x + ay$ on any PDE. It only works when x and y do not appear explicitly. For $pq = xz$ the explicit x destroys the translation invariance, and the substitution does not reduce to an ODE in u alone.

Worked Example

(1) $p(1+q) = qz$**.** With $p = z'$, $q = az'$: $z'(1+az') = az'\,z \Rightarrow 1 + az' = az$. So $\dfrac{dz}{az-1} = \dfrac{du}{a}$, giving $\ln(az-1) = u + \text{const}$:

$$az - 1 = Ce^{x+ay} \iff z = \frac{1 + Ce^{x+ay}}{a}.$$

(2) $pq = z$**.** $az'^2 = z \Rightarrow \dfrac{dz}{\sqrt{z}} = \dfrac{du}{\sqrt{a}} \Rightarrow 2\sqrt{z} = \dfrac{u+b}{\sqrt{a}}$, i.e. $4az = (x + ay + b)^2$.

(3) $z^2(p^2 + q^2 + 1) = 1$**.** $z^2\big[(1+a^2)z'^2 + 1\big] = 1 \Rightarrow \dfrac{z\,dz}{\sqrt{1-z^2}} = \dfrac{du}{\sqrt{1+a^2}}$. Integrating, $-\sqrt{1-z^2} = \dfrac{u+b}{\sqrt{1+a^2}}$, so

$$(x + ay + b)^2 = (1 + a^2)(1 - z^2),$$

a family of ellipsoids/spheres.

28 Equations of the Form $F(x,p) = G(y,q)$

Definition

Method. Set both sides equal to a constant a: $F(x,p) = a = G(y,q)$. Solve $p = f(x,a)$ and $q = g(y,a)$, then integrate $dz = p\,dx + q\,dy$:

$$z = \int f(x,a)\,dx + \int g(y,a)\,dy + b.$$

WHY this is true

The left side involves only x (and p), the right only y (and q). We look for solutions in which p depends on x alone and q on y alone. Then the equation says "a function of x equals a function of y for *all* x and y". Vary x while keeping y fixed: the right side does not move, so the left side cannot either. Hence both are the same constant a. This is separation of variables applied to the *slopes* p, q rather than to z.

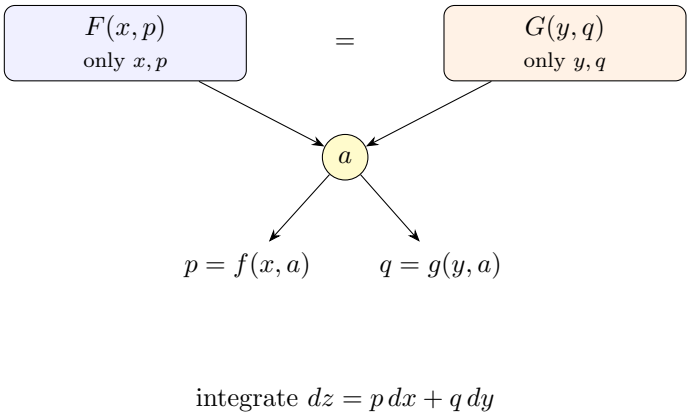


Figure 16: Both sides of the equation must equal the same separation constant a, which then splits the PDE into two one-variable problems.

Fixing a Common Misconception

"$p+q = x+y$ can't be separated because x and y are mixed." Rearrange first: $p - x = y - q$ has the form $F(x,p) = G(y,q)$. Then $p = x + a$, $q = y - a$, and $z = \frac{x^2}{2} + ax + \frac{y^2}{2} - ay + b$. Always try to move all x, p-terms to one side and all y, q-terms to the other.

Worked Example

(1) $p^2 + q^2 = x + y$**.** $p^2 - x = y - q^2 = a$: $p = \sqrt{x+a}$, $q = \sqrt{y-a}$, so $z = \frac{2}{3}(x+a)^{3/2} + \frac{2}{3}(y-a)^{3/2} + b$.

(2) $p\,e^y = q\,e^x$**.** $pe^{-x} = qe^{-y} = a$: $p = ae^x$, $q = ae^y$, so $z = a(e^x + e^y) + b$.

(3) $p^2y(1+x^2) = qx^2$**.** Divide both sides by x^2y: $\dfrac{p^2(1+x^2)}{x^2} = \dfrac{q}{y} = a$, so $p = \dfrac{\sqrt{a}\,x}{\sqrt{1+x^2}}$, $q = ay$ and $z = \sqrt{a}\sqrt{1+x^2} + \dfrac{ay^2}{2} + b$. **Check:** $p^2y(1+x^2) = \dfrac{ax^2}{1+x^2}y(1+x^2) = ax^2y = qx^2$. ✓

29 Clairaut's Form and the Singular Solution

Definition

$$z = px + qy + f(p, q).$$

Complete integral: replace p, q by constants: $z = ax + by + f(a, b)$. **Singular solution:** differentiate the complete integral with respect to a and b: $x + f_a = 0,\ y + f_b = 0$; eliminate a, b.

WHY this is true

Why the complete integral is this simple. If $z = ax + by + f(a, b)$ then $p = a,\ q = b$, and substituting gives $z = ax + by + f(a, b)$ — an identity. So replacing p, q by constants *is* a solution.

What the singular solution is. The complete integral is a two-parameter family of planes. Their *envelope* — the surface tangent to every plane of the family — is the singular solution. Its tangent plane at each point is one of the family's planes, so it satisfies the same PDE, yet it cannot be obtained by fixing a, b. The conditions $\partial/\partial a = \partial/\partial b = 0$ are exactly the envelope conditions. The one-variable picture below shows the same thing: lines $y = ax + a^2$ (Clairaut's ODE $y = xy' + y'^2$) and their envelope, the parabola $y = -x^2/4$.

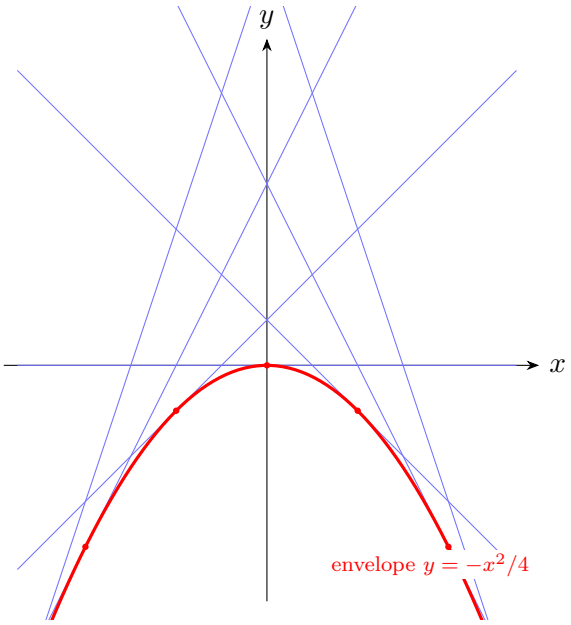


Figure 17: One-variable analogue: the lines $y = ax + a^2$ (blue) all touch the parabola (red). The parabola is not any one of the lines, yet at each point it has the slope of the line touching there, so it solves the same equation: it is the singular solution.

Fixing a Common Misconception

"$z = -xy$ solves $z = px+qy+pq$ by putting $a = -y,\ b = -x$ in the complete integral." The constants a, b must stay constant; substituting functions is not allowed. $z = -xy$ *is* a solution ($p = -y,\ q = -x$: $px + qy + pq = -xy - xy + xy = -xy$) but it is the singular solution — the envelope — obtained by $x + b = 0,\ y + a = 0$.

Worked Example

(1) $z = px+qy+p^2+q^2$**.** Complete: $z = ax+by+a^2+b^2$. Differentiating: $x+2a = 0$, $y+2b = 0$, so $a = -\frac{x}{2}$, $b = -\frac{y}{2}$, and $z = -\dfrac{x^2+y^2}{2} + \dfrac{x^2+y^2}{4} = -\dfrac{x^2+y^2}{4}$, i.e. $4z = -(x^2+y^2)$.

(2) $z = px + qy + \sqrt{1+p^2+q^2}$**.** Complete: $z = ax + by + s$ with $s = \sqrt{1+a^2+b^2}$. Then $x = -\dfrac{a}{s}$, $y = -\dfrac{b}{s}$, $z = -\dfrac{a^2+b^2}{s} + s = \dfrac{1}{s}$, so $x^2+y^2+z^2 = \dfrac{a^2+b^2+1}{s^2} = 1$: the singular solution is the unit sphere (tangent planes of a sphere are exactly this family).

(3) Reduce to Clairaut first. $(p+q)(z-xp-yq) = 1$ rearranges to $z = px + qy + \dfrac{1}{p+q}$: Clairaut form with $f = \dfrac{1}{p+q}$. Complete integral: $z = ax + by + \dfrac{1}{a+b}$.

30 Lagrange's Linear Equation $Pp + Qq = R$

Definition

Here P, Q, R are functions of x, y, z. The **auxiliary (subsidiary) equations** are

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}.$$

If $u(x, y, z) = c_1$ and $v(x, y, z) = c_2$ are two independent solutions, the **general solution** is $\phi(u, v) = 0$ (ϕ arbitrary), or equivalently $v = f(u)$.

WHY this is true

Step 1: what the PDE says geometrically. A surface $z = z(x, y)$ has normal $(p, q, -1)$. The equation $Pp+Qq = R$ says $(P, Q, R)\cdot(p, q, -1) = 0$: the vector (P, Q, R) is *perpendicular to the normal*, so it is *tangent* to the surface at every point. An integral surface is therefore a surface to which the vector field (P, Q, R) is everywhere tangent.

Step 2: such surfaces are woven from characteristic curves. Follow a curve with $\frac{dx}{dt} = P$, $\frac{dy}{dt} = Q$ starting on the surface. By the chain rule along the surface, $\frac{dz}{dt} = p\frac{dx}{dt} + q\frac{dy}{dt} = pP + qQ = R$. So the curve has tangent (P, Q, R) and stays on the surface: eliminating t gives the auxiliary equations. The surface is a union of such curves.

Step 3: the general solution. The curves are $u = c_1$, $v = c_2$ (two independent first integrals). A surface made of a one-parameter family of them is obtained by linking the parameters through any relation $\phi(c_1, c_2) = 0$ — hence $\phi(u, v) = 0$, with ϕ arbitrary. This also matches Section 25, where eliminating an arbitrary ϕ from $\phi(u, v) = 0$ produced exactly this PDE.

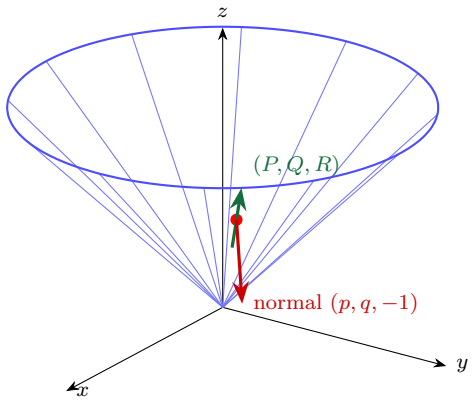


Figure 18: An integral surface of $xp + yq = z$: the cone $z = \sqrt{x^2 + y^2}$, built from straight lines through the origin. At each point $(P, Q, R) = (x, y, z)$ points *along* the ray (tangent, green) and is perpendicular to the surface normal $(p, q, -1)$ (red).

Solving the auxiliary equations

WHY this is true

Two techniques. (i) **Grouping**: if two ratios involve only two variables, integrate directly, e.g. $\frac{dx}{P} = \frac{dy}{Q}$. (ii) **Multipliers**: for any functions l, m, n, equal ratios satisfy

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} = \frac{l\,dx + m\,dy + n\,dz}{lP + mQ + nR}.$$

(If every ratio equals κ, then $dx = \kappa P$, etc., so the numerator is $\kappa(lP + mQ + nR)$.) Choose l, m, n so that $lP + mQ + nR = 0$; then $l\,dx + m\,dy + n\,dz = 0$, and if that is an exact differential you have a first integral.

Worked Example

(1) $xp+yq = z$**.** $\frac{dx}{x} = \frac{dy}{y} = \frac{dz}{z}$ gives $\dfrac{y}{x} = c_1$, $\dfrac{z}{x} = c_2$. Solution: $\phi\left(\frac{y}{x}, \frac{z}{x}\right) = 0$, i.e. $z = x\,f(y/x)$ — cones, as in the figure.

(2) $yz\,p + zx\,q = xy$**.** $\frac{dx}{yz} = \frac{dy}{zx} = \frac{dz}{xy}$. From the first two, $x\,dx = y\,dy$; from the last two, $y\,dy = z\,dz$. So $x^2 - y^2 = c_1$, $y^2 - z^2 = c_2$: $\phi(x^2 - y^2,\ y^2 - z^2) = 0$.

(3) $(y - z)p + (z - x)q = x - y$ **(multipliers).** With $(l, m, n) = (1, 1, 1)$: $lP + mQ + nR = (y-z)+(z-x)+(x-y) = 0$, so $d(x+y+z) = 0$, giving $x+y+z = c_1$. With $(l, m, n) = (x, y, z)$: $x(y-z)+y(z-x)+z(x-y) = 0$, so $x\,dx+y\,dy+z\,dz = 0$, giving $x^2+y^2+z^2 = c_2$. Solution:

$$\phi(x + y + z,\ x^2 + y^2 + z^2) = 0.$$

This is exactly the surface family from which Section 25 formed this PDE.

(4) $(y + z)p + (z + x)q = x + y$**.** Taking differences: $\frac{d(x-y)}{-(x-y)} = \frac{d(y-z)}{-(y-z)}$, so $\dfrac{x - y}{y - z} = c_1$. Using $(1, 1, 1)$ (numerator $d(x + y + z)$, denominator $2(x + y + z)$) against $\frac{d(x-y)}{-(x-y)}$: $\dfrac{d(x + y + z)}{2(x + y + z)} = -\dfrac{d(x - y)}{x - y}$, so $(x - y)^2(x + y + z) = c_2$. Solution: $\phi\left(\dfrac{x - y}{y - z},\ (x - y)^2(x + y + z)\right) = 0$.

Fixing a Common Misconception

"The general solution is $u + v = c$." The general solution is $\phi(u, v) = 0$ with ϕ *arbitrary*. Writing $u + v = c$ picks one specific ϕ and gives only a one-parameter family — far fewer solutions than the PDE has.

31 Linear First-Order PDEs by Separation of Variables

Definition

Method. Assume $u = X(x)Y(y)$ (or $X(x)T(t)$). Substitute and rearrange so that each side depends on one variable only; set both equal to a separation constant k. Solve the two ODEs, form the product, and use the given condition to fix k and the multiplicative constant.

WHY this is true

Why a constant? After substitution one side is a function of x only and the other a function of y (or t) only, and they are equal for all values of both variables — so each equals the same constant k (as in Section 28). Why does the product form lose nothing? For a linear PDE, sums of solutions are solutions (*superposition*), so if the given condition is a sum of exponentials, we solve each term separately and add.

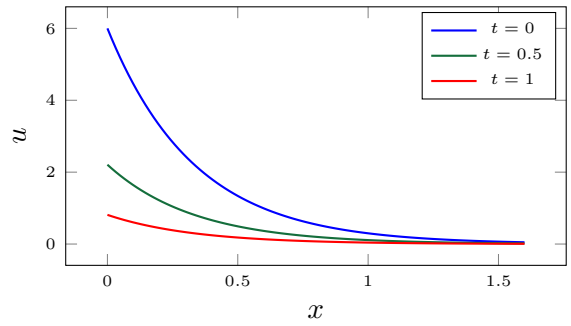


Figure 19: The solution $u = 6e^{-3x-2t}$ of $u_x = 2u_t + u$ with $u(x,0) = 6e^{-3x}$: the initial profile (blue) decays uniformly in time, since u is multiplied by e^{-2t}.

Worked Example

(1) $u_x = 2u_t + u$, $u(x,0) = 6e^{-3x}$. $u = XT$: $X'T = 2XT' + XT \Rightarrow \dfrac{X'}{X} = 2\dfrac{T'}{T} + 1 = k$. So $X = e^{kx}$, $T = e^{(k-1)t/2}$, $u = Ce^{kx+(k-1)t/2}$. At $t = 0$: $Ce^{kx} = 6e^{-3x}$, so $C = 6$, $k = -3$:

$$u = 6e^{-3x-2t}.$$

(2) $3u_x + 2u_y = 0$, $u(x,0) = 4e^{-x}$. $\dfrac{X'}{X} = -\dfrac{2}{3}\dfrac{Y'}{Y} = k$: $u = Ce^{kx-3ky/2}$. Condition: $C = 4$, $k = -1$, so $u = 4e^{-x+3y/2}$.

(3) Superposition. If instead $u(x,0) = 6e^{-3x} + 4e^{-5x}$ in (1), each term has its own k (-3 and -5): $u = 6e^{-3x-2t} + 4e^{-5x-3t}$.

Connection — where this reappears

Example (2) is also a Lagrange equation ($P = 3$, $Q = 2$, $R = 0$): the auxiliary equations give $2x - 3y = c_1$, $u = c_2$, so $u = f(2x - 3y)$. The separated answer $4e^{-x+3y/2} = 4e^{-(2x-3y)/2}$ is exactly of this form with $f(s) = 4e^{-s/2}$. Two different methods, one answer — and Lagrange's method shows that *every* solution (not just exponential ones) is a function of $2x - 3y$.

Final Exam-Prep Checklist

- ☐ Can you explain, out loud, why ∇f is a vector even though f is a scalar — without looking at Section 2?
- ☐ Can you compute gradient, divergence, and curl from memory, without hesitating on the **j**-sign in curl?
- ☐ Can you find the maximum directional derivative and its direction at a point, from a bare function?
- ☐ Can you find a scalar potential φ for a given irrotational $\mathbf{F}$, showing all three matching steps?
- ☐ Can you state, in one sentence each, what Green's, Stokes', and Gauss's theorems convert into what, and reproduce the cancellation argument for at least one of them?
- ☐ Can you verify the Divergence Theorem on a cube (both sides, independently) without looking anything up?
- ☐ Can you sketch a region and flip the order of a double integral without algebra mistakes?
- ☐ Do you know, on sight, which coordinate system a given region calls for, and can you state why the Jacobian factor (r, or $\rho^2 \sin\varphi$) is what it is?
- ☐ Can you solve the auxiliary equation and write CF correctly for all three root cases?
- ☐ Can you recognize a resonance (failure) case immediately, and know whether to multiply by x or x^2?

Work through every worked example in this document a second time, from a blank page, before your exam — reproducing a solution (and a WHY argument) without looking is the real test of understanding, not whether the explanation made sense while reading it.

Part II — Quick-Reference Summary

Module 1	
Total differential	$dz = f_x\,dx + f_y\,dy$ (linear estimate of Δz; valid for small steps)
Jacobian	$\dfrac{\partial(x,y)}{\partial(u,v)} = x_u y_v - x_v y_u$; area factor; $J \cdot J^{-1} = 1$; $J \equiv 0 \Leftrightarrow$ dependent
Taylor (2 var.)	$f + (hf_x + kf_y) + \frac{1}{2!}(h^2 f_{xx} + 2hkf_{xy} + k^2 f_{yy}) + \cdots$
Max/min test	$f_x = f_y = 0$; $rt - s^2 > 0$: min ($r > 0$) / max ($r < 0$); < 0: saddle; $= 0$: inconclusive
Lagrange	$\nabla f = \lambda \nabla g,\ g = 0$; compare candidate values
Module 4 (continued)	
Variation of parameters	$y_p = -y_1 \int \frac{y_2 R}{W} + y_2 \int \frac{y_1 R}{W}$, $W = y_1 y_2' - y_1' y_2$
Cauchy–Euler	$x = e^t$; $m(m-1) + am + b = 0$; x^{m_1}, x^{m_2} / $(C_1 + C_2 \ln x)x^m$ / $x^\alpha \cos(\beta \ln x)$
Cauchy–Legendre	$ax + b = e^t$; $(ax+b)y' = aDy$, $(ax+b)^2 y'' = a^2 D(D-1)y$
Spring/circuit	$mx'' + cx' + kx = F$; $Lq'' + Rq' + q/C = E$; $c^2 - 4mk$: over / critical / under
Module 5	
$F(p,q) = 0$	$z = ax + \varphi(a)y + c$ with $F(a, \varphi(a)) = 0$
$F(z,p,q) = 0$	$u = x + ay$, $p = z'$, $q = az'$, solve ODE
$F(x,p) = G(y,q)$	both sides $= a$; $dz = p\,dx + q\,dy$
Clairaut	$z = px + qy + f(p,q) \Rightarrow z = ax + by + f(a,b)$; singular: $\partial_a = \partial_b = 0$
Lagrange	$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$; $\phi(u,v) = 0$; use grouping or multipliers
Separation of variables	$u = XY$; $X'/X = \cdots = k$; fix k from the condition; superpose

Part II — Exam-Prep Checklist

- ☐ Can you explain, with a picture, what f_x means, and give an example where f_x, f_y exist but f is discontinuous?
- ☐ Can you explain why dz approximates Δz, and use it for an error-propagation problem?
- ☐ Can you state why $|J|$ is an area factor, and derive the polar (r) and spherical $(\rho^2 \sin\varphi)$ Jacobians?
- ☐ Can you derive the second-degree Taylor terms by restricting f to a line?
- ☐ Can you derive the $rt - s^2$ test by completing the square, and handle the inconclusive case?
- ☐ Can you argue why $\nabla f \parallel \nabla g$ at a constrained extremum, and decide max versus min among candidates?
- ☐ Can you derive the variation-of-parameters formula, including the extra condition $u_1' y_1 + u_2' y_2 = 0$?
- ☐ Can you reduce Cauchy–Euler and Cauchy–Legendre equations to constant coefficients, including resonant right-hand sides?
- ☐ Can you set up and classify a spring or LCR problem, and recognise resonance?
- ☐ Can you form a PDE by eliminating constants and by eliminating functions?
- ☐ Given a first-order PDE, can you identify its type $(F(p,q)$, $F(z,p,q)$, separable, Clairaut, Lagrange) and apply the right method?
- ☐ Can you explain, geometrically, why Lagrange's auxiliary equations work, and use multipliers?
- ☐ Can you find the singular solution of a Clairaut equation and say what it is geometrically?